

Operations Management

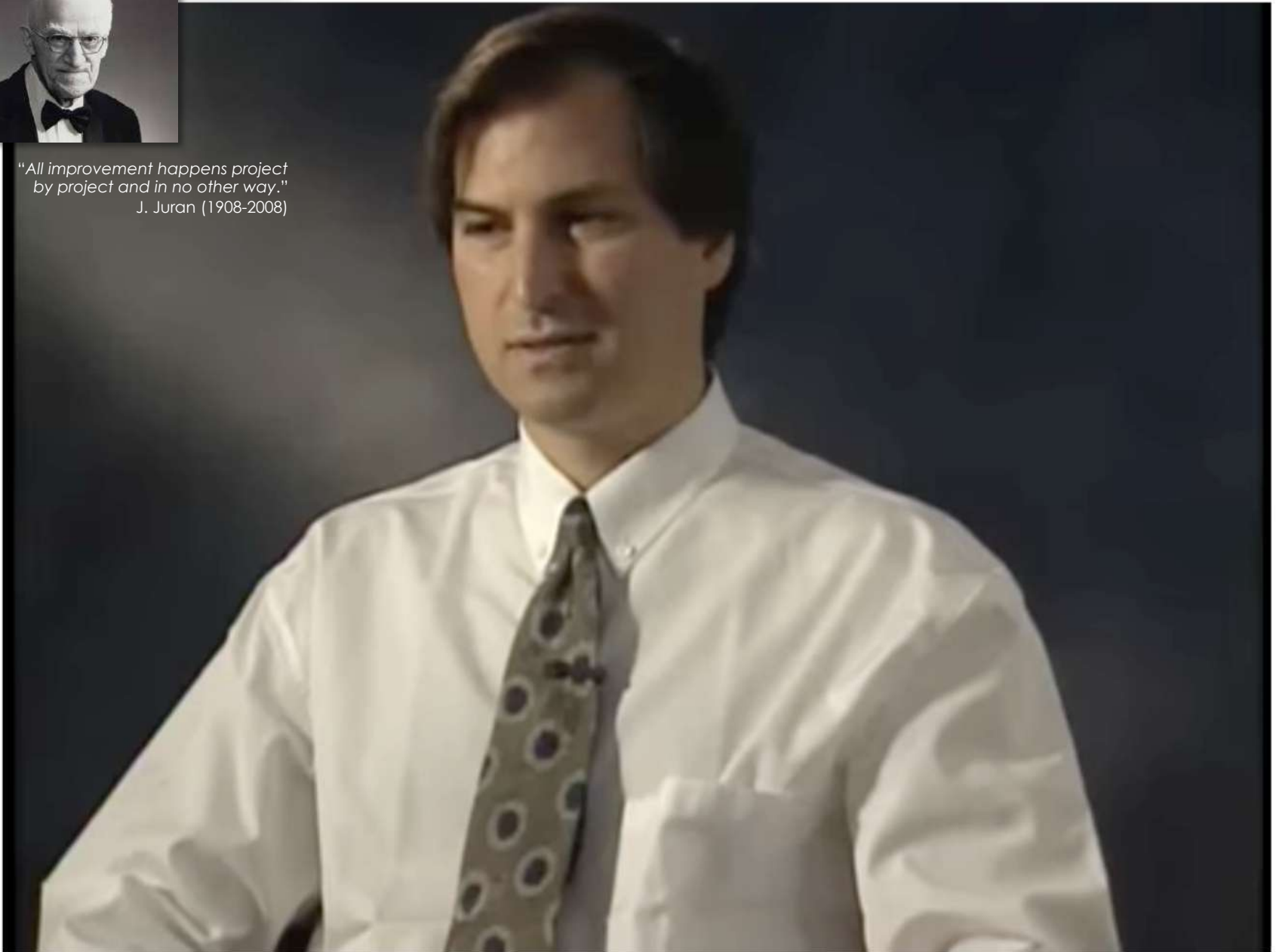
Session 5: Process Improvement and Lean Operations

Outline:

- Statistical Process Control
 - Process Capabilities
 - X-Charts: controlling for Type I and Type II errors
 - Correcting without overcorrecting
 - The limits of SPC: The risk equation
- A Modern View: Toyota Production System & TQM
 - Just-in-time production
 - Andon controls
 - Employee empowerment
 - The House of Quality: Kaizen, Heijunka, Jidoka, Poka-Yoke
- Lean Operations
 - The six-sigma philosophy of continuous improvement
 - Prevention is more effective than correction

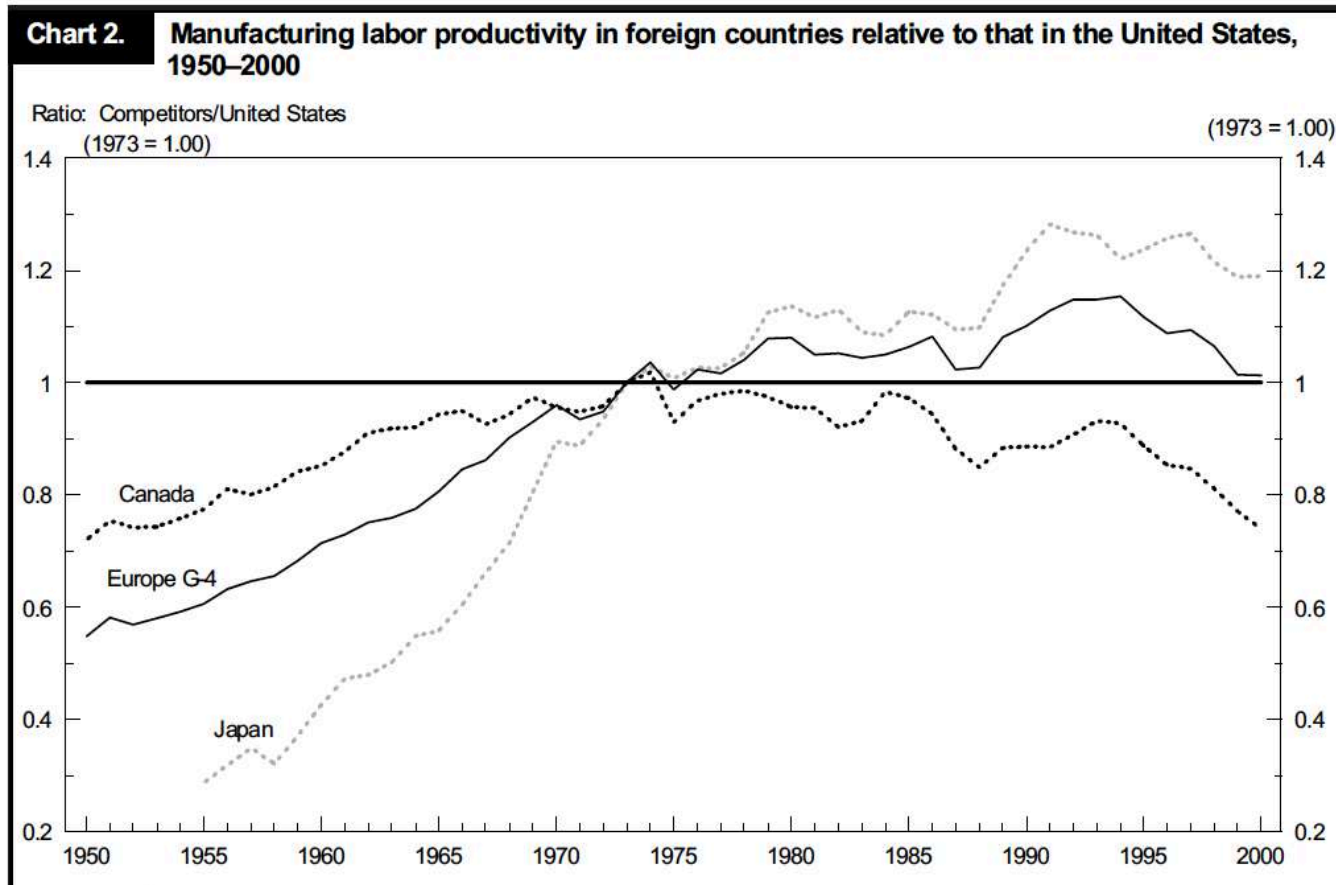


*"All improvement happens project
by project and in no other way."*
J. Juran (1908-2008)



Steve Jobs on Joseph Juran and Quality ([available on YouTube](#))

The Productivity Race



Conformance to Specifications
(Product-Based)



Quality Assurance
(Customer-Based)



Total Quality Management
(Process-Based and Organization-wide)

Traditional Quality Control

From Defects to Causes to Corrective Action



“Inspection is too late. The quality, good or bad, is already in the product.”
— W.E. Deming (1900-1993)

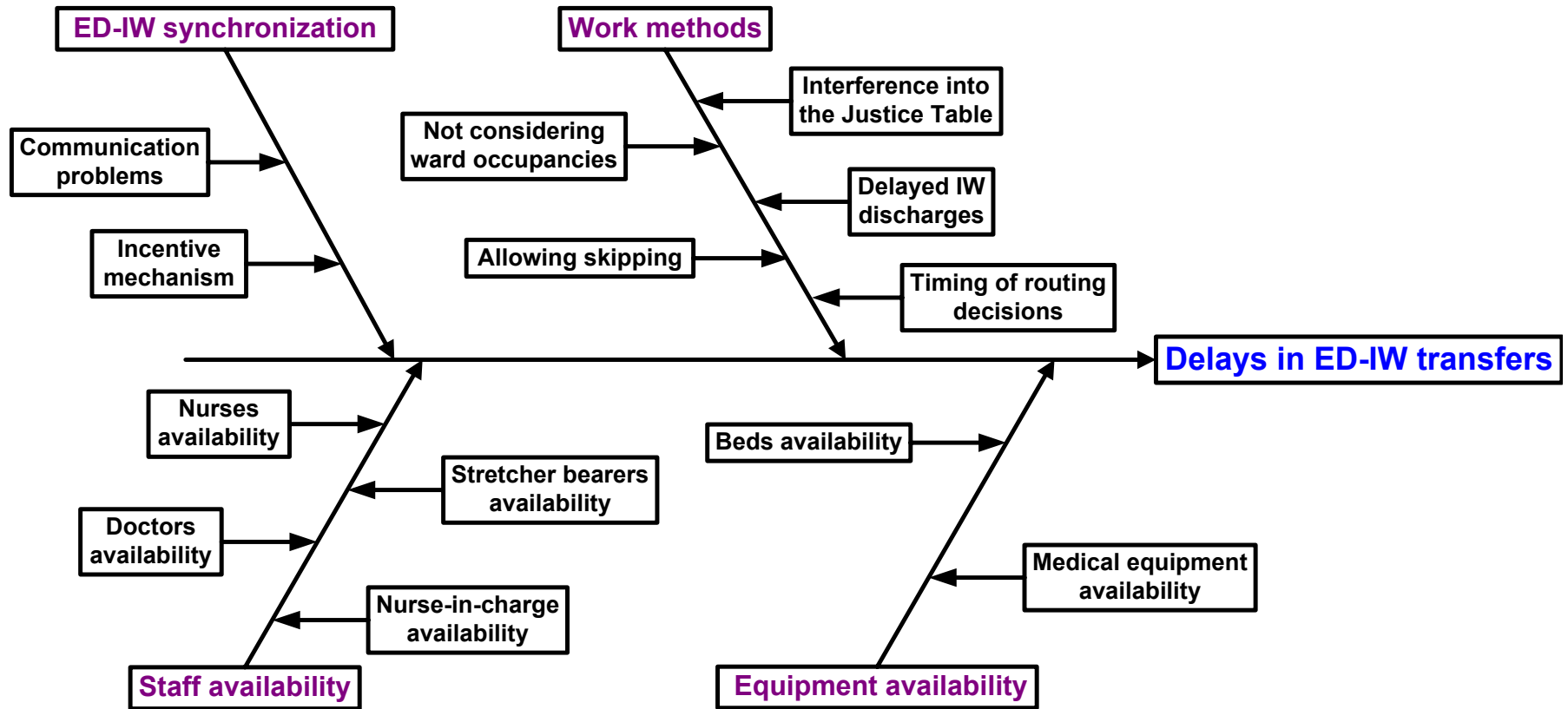
- GOAL: keep the process **on target** by detecting **defects/deviations** and eliminating their underlying **causes**.

- LOGIC: treat quality as a **closed-loop feedback system**:

observe → diagnose → intervene → verify.

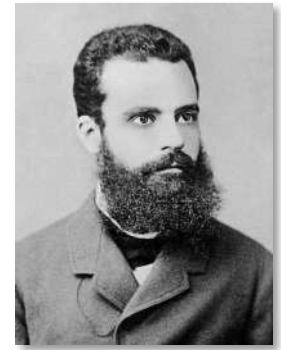
- DIAGNOSE: use structured tools to connect **“what went wrong”** (defect type, time, location, product, operator, supplier lot) to **“why it happened”** (process conditions, methods, machines, materials, environment).
- PRIORITIZE: focus effort where it matters most, separate the **vital few** drivers from the **trivial many**.
- CORRECT & PREVENT: implement **corrective actions** (process changes, standard work, training, maintenance, error-proofing, supplier fixes) and confirm the improvement is sustained.

Diagnostics Tools: Fishbone Diagram



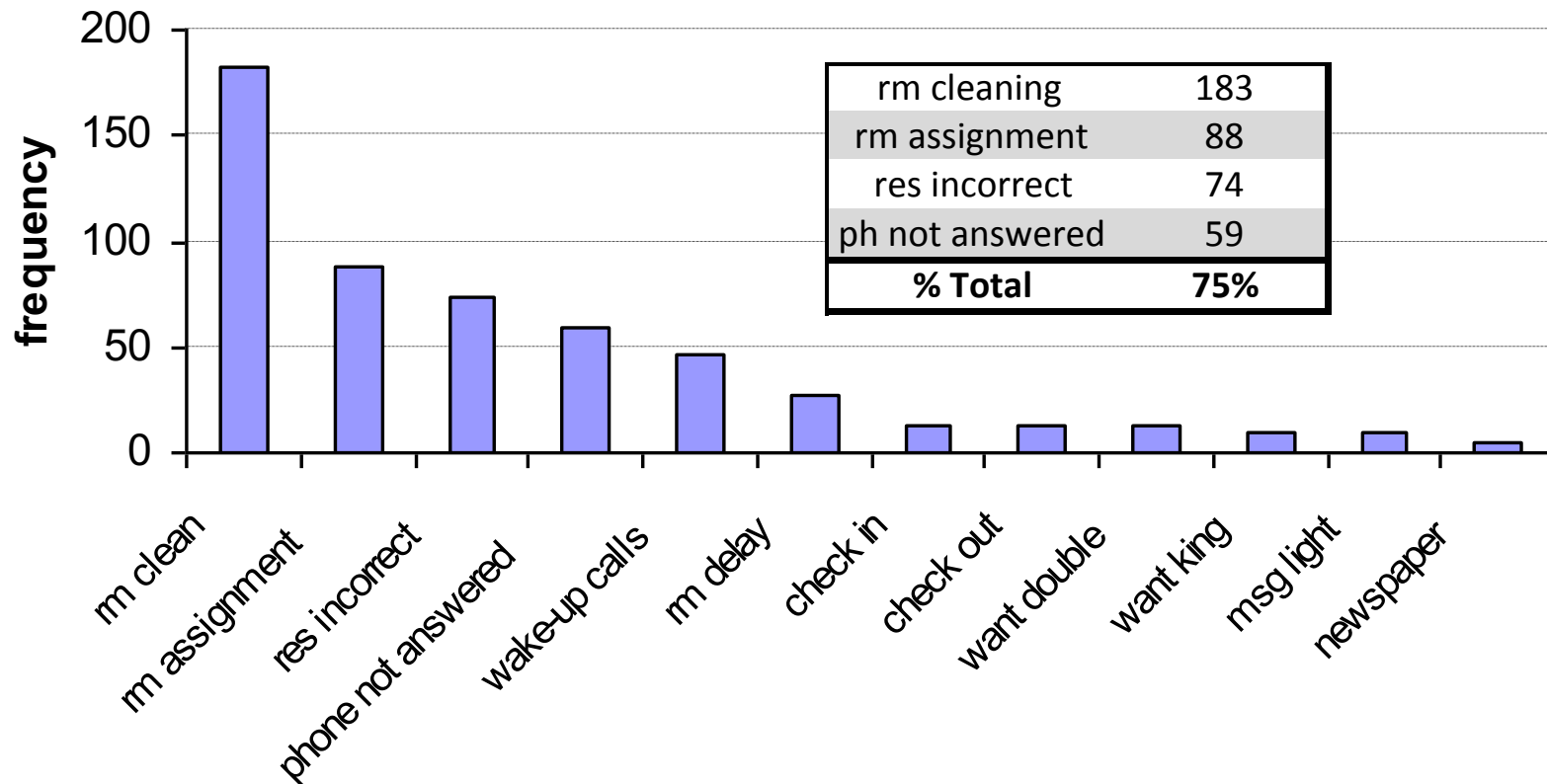
Source: Armony et al, Patient flow in hospitals: a data-based queueing-science perspective, 2011

Diagnostics Tools: Pareto Chart

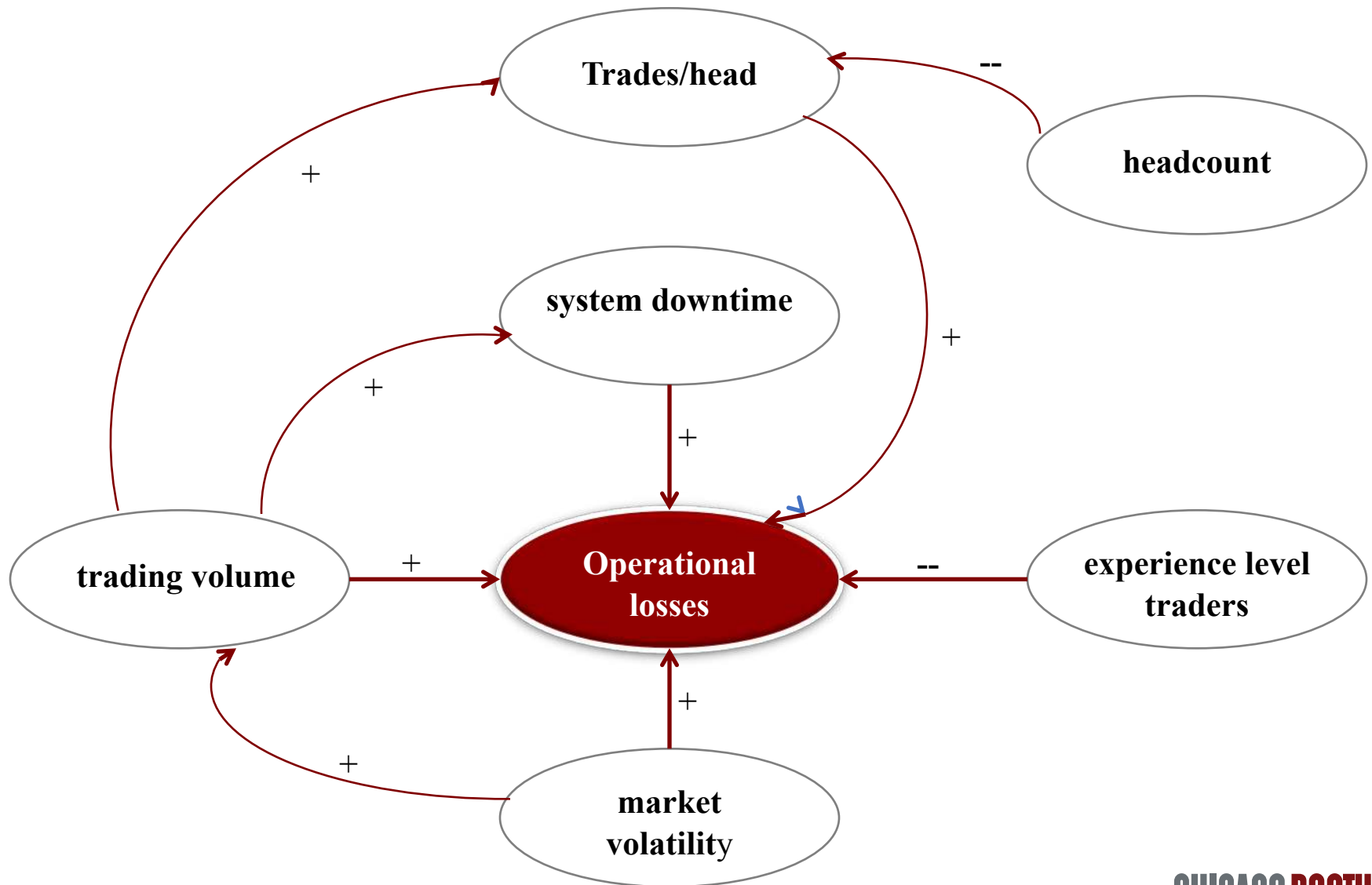


80% of the wealth held
by 20% of the population.
Vilfredo Pareto (1848–1923)

All Defects by Keywords



Diagnostics Tools: Influence Diagram



Diagnostics Tools: Linear Regressions

‘Loss’ vs. ‘Trades by Headcount’

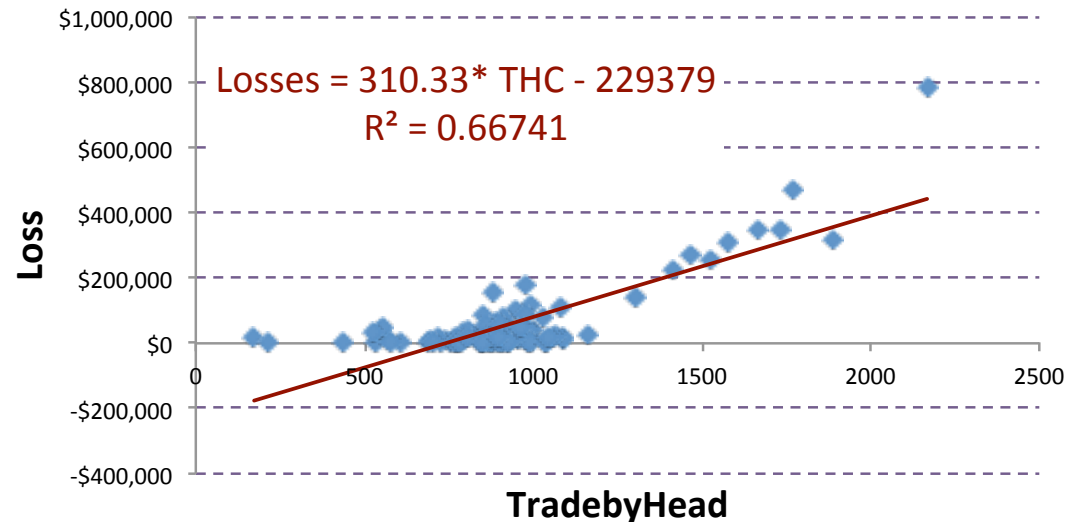
SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.816952324
R Square	0.6674111
Adjusted R Square	0.664085211
Standard Error	65627.6975
Observations	102

ANOVA

	df	SS	MS	F	Significance F
Regression	1	8.64291E+11	8.64E+11	200.67149	1.20767E-25
Residual	100	4.30699E+11	4.31E+09		
Total	101	1.29499E+12			

	Coefficients	Standard Error	t Stat	P-value	Lower 95%
Intercept	-229378.9394	21368.59137	-10.7344	2.44365E-18	-271773.616
TradebyHead	310.3279677	21.90675644	14.16586	1.20767E-25	266.8655869



Statistical Process Control

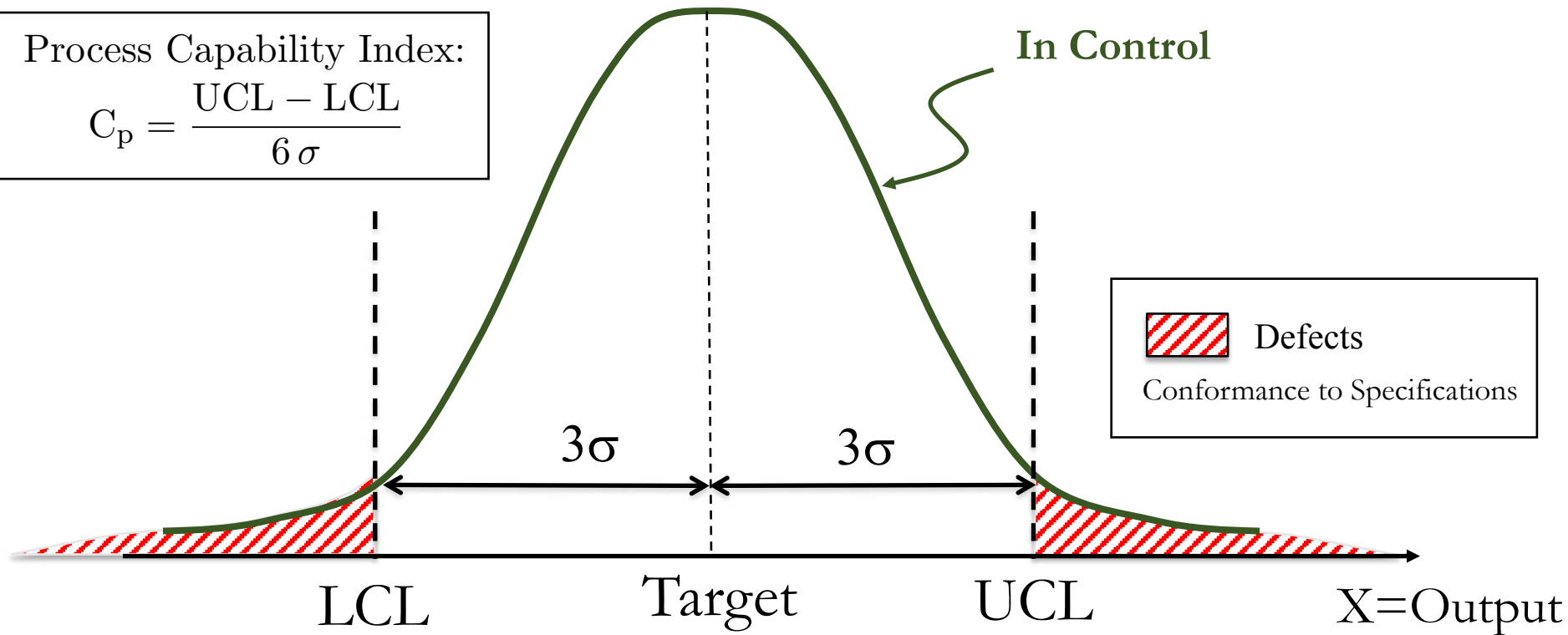
In God we
trust, all
others bring
data.

—William E. Deming



Process Capability Index:

$$C_p = \frac{UCL - LCL}{6\sigma}$$



Examples:

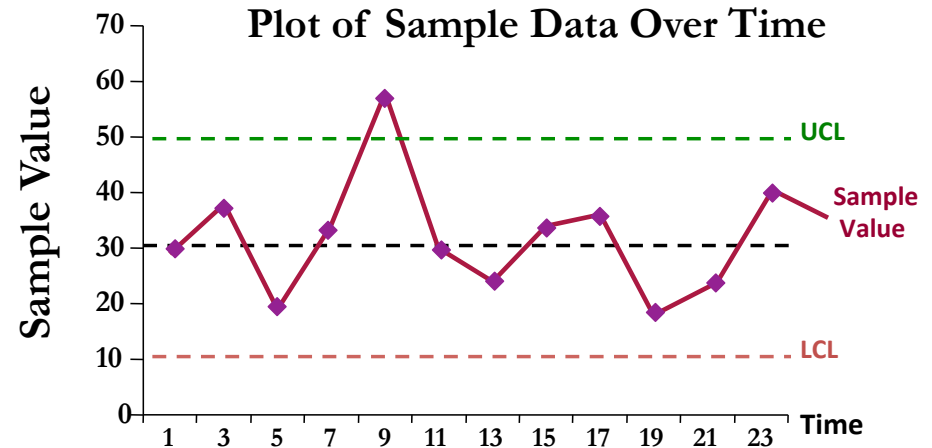
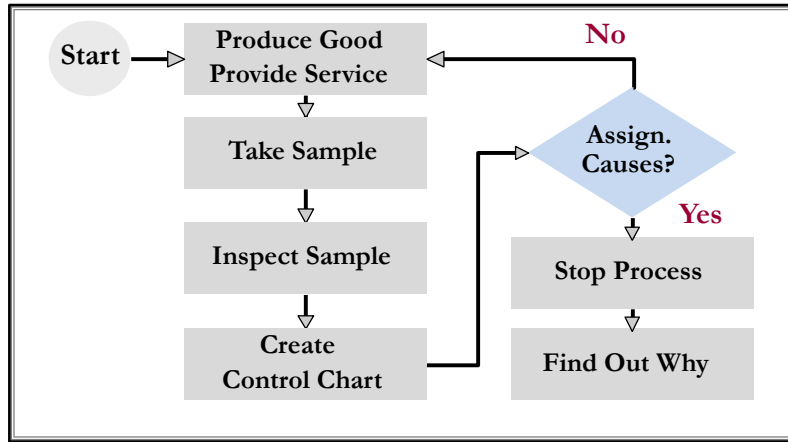
Manufacturing: Fill volume in bottling/canning

Healthcare: Lab analyzer result for a control sample

Services: Call-center average response time

Control Charts or X-Chart

Reduce the variation and center the process ... *Customers feel the variance not the mean!*



	In Control	Not in Control
Assume Process is OK	Random Variation	Type II Error
Take Corrective Action	Type I Error	Assignable Variation

Type I Error: Probability that a process operating “In Control” produces a unit outside the control limits

Type II Error: Probability that a process operating “Out of Control” produces a unit inside the control limits

Statistical Process Control Chart

Central line-associated Blood Stream Infections (BSI)

Characteristic mean = 3.64 BSI per 1000 central line-days

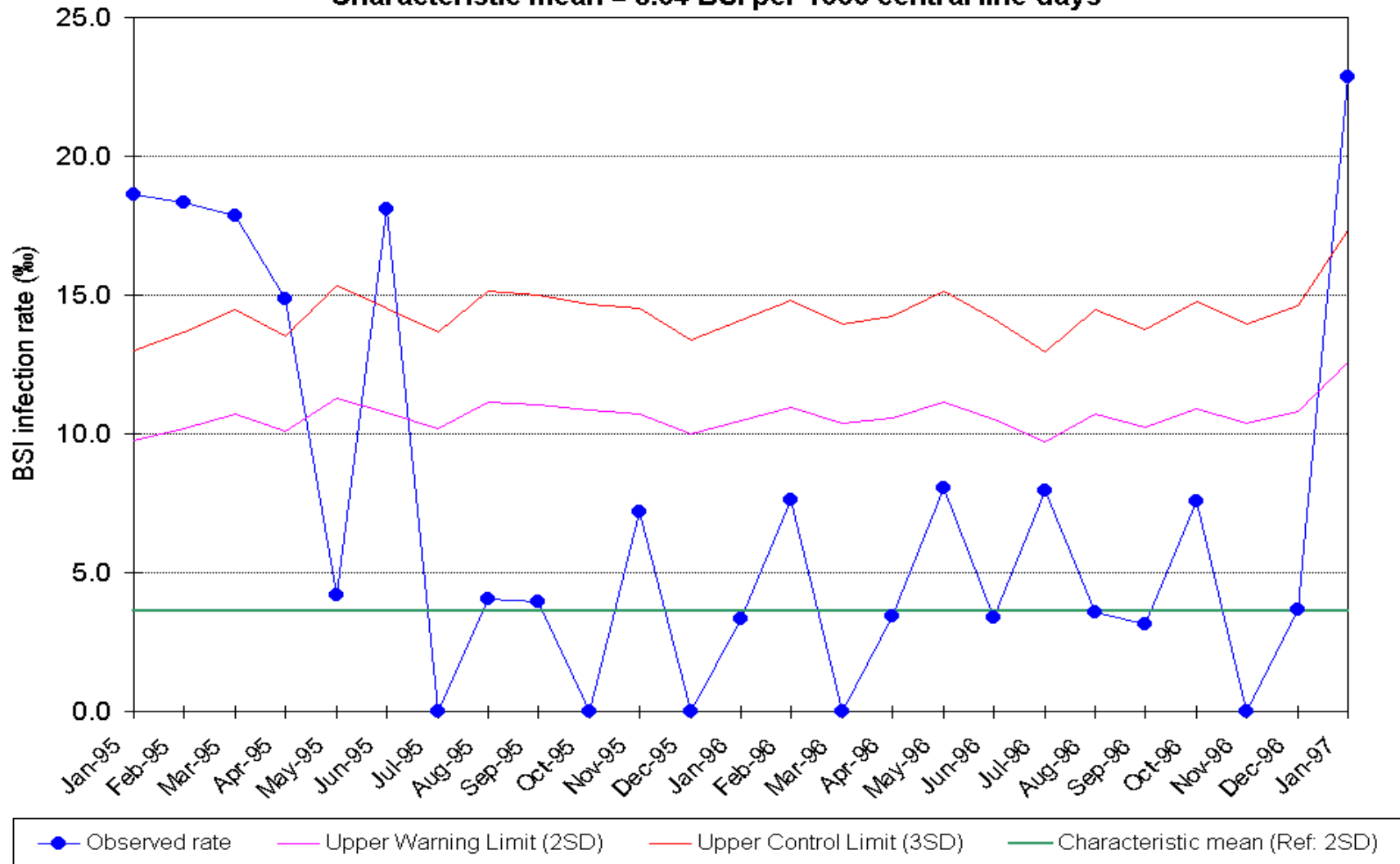


Figure 3. This is the endemic curve corresponding to period between January 1995 and January 1997. As the model A predicted, an outbreak occurred in January 1997.

Example

A normally distributed quality characteristic is monitored using statistical control. If the process is in control then this quality attribute has a mean of 50 and standard deviation of 10. The process has a tendency to go out of control in which case the mean of the quality attribute shifts from 50 to 45. On the other hand, the standard deviation remains at 10.

A decision has been made to stop the process and take corrective actions if there is an output that is two or more standard deviations away from the target mean.

a) What is the probability of a type I error, that is, that a particular sample falls outside the limits but the process is operating in control?

$$\text{Upper Control Limit (UCL)} = 50 + 2 * 10 = 70$$

$$\text{Lower Control Limit (LCL)} = 50 - 2 * 10 = 30$$

For a 2-sigma process the yield is 95.4%.

Type-I Error = 1-yield = 4.6%

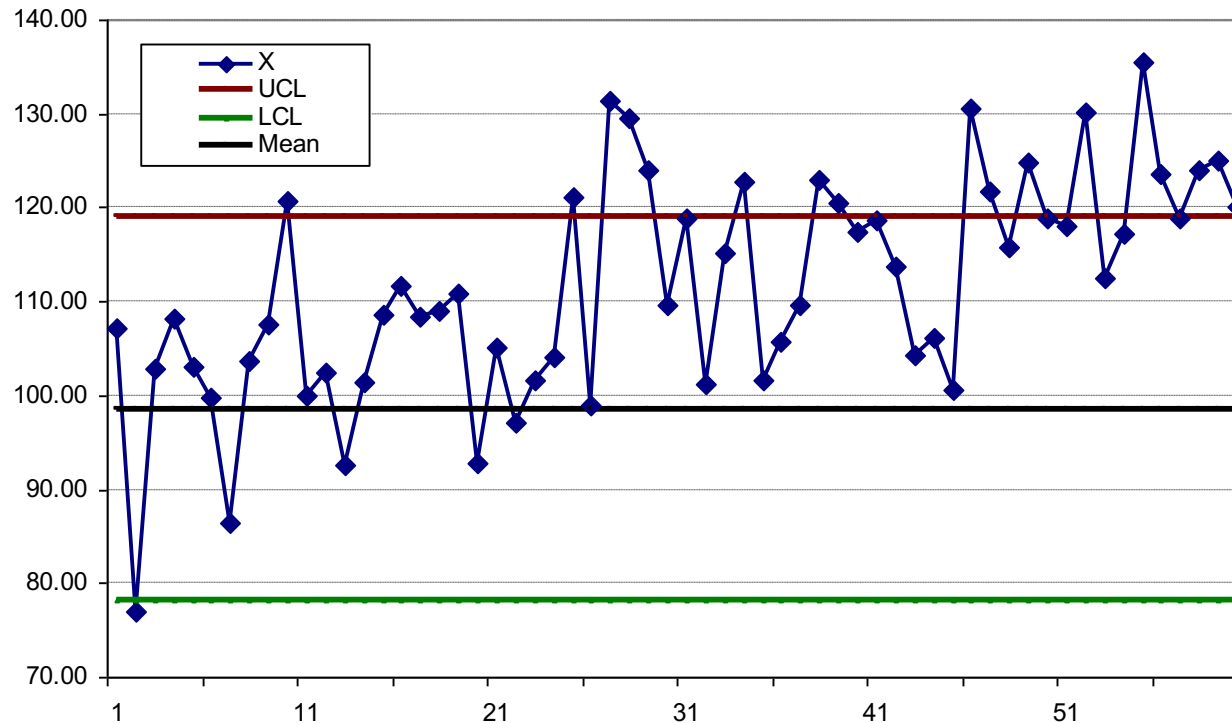
b) What is the probability of a type-II error, that is, that a particular sample is within the control limits given that the process is operating out of control?

$$\text{Type-II Error} = \text{Prob}(30 \leq \text{Sample} \leq 70) = \text{Prob}\left(\frac{30-45}{10} \leq Z \leq \frac{70-45}{10}\right) = \text{Prob}(-1.5 \leq Z \leq 2.5)$$

$$\text{Type-II Error} = 0.994 - 0.0668 = 92.7\%$$

Example

The following is an X-chart for a process with a target mean of 98 and a standard deviation of 20. The upper and lower control limits are set at 118 and 78, respectively.



Based on this information, in which sample point you would stop the process for the first time?

When is Corrective Action Required?

- When there is sufficient evidence that some “assignable cause” has affected the process !

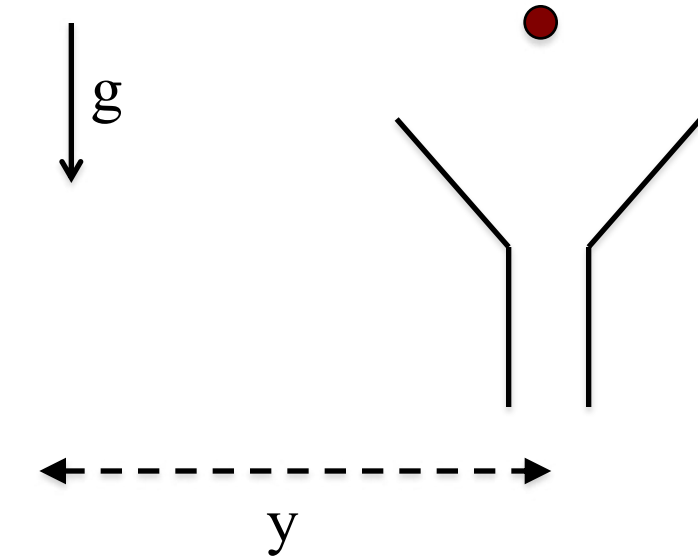
- Four indications that a process is “out of control”.
 - ✓ One point falls outside the control limits
 - ✓ Seven points in a row all on one side of the center line
 - ✓ A run of seven points in a row going up, or a run of seven points in a row going down
 - ✓ Cycles or other non-random patterns

- If any one of these things happens, you should stop the process and inspect (quality control).

Deming's Funnel Experiment



W.E. Deming (1900-1993)

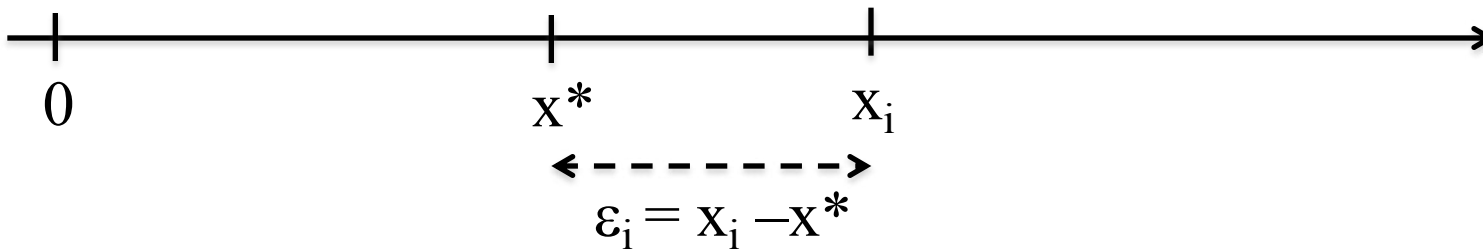


- 1st Idea: Adjust

$$y_{\text{new}} = y_{\text{old}} - \epsilon_i$$

- 2nd Idea: Do not adjust

$$y_{\text{new}} = x^*$$

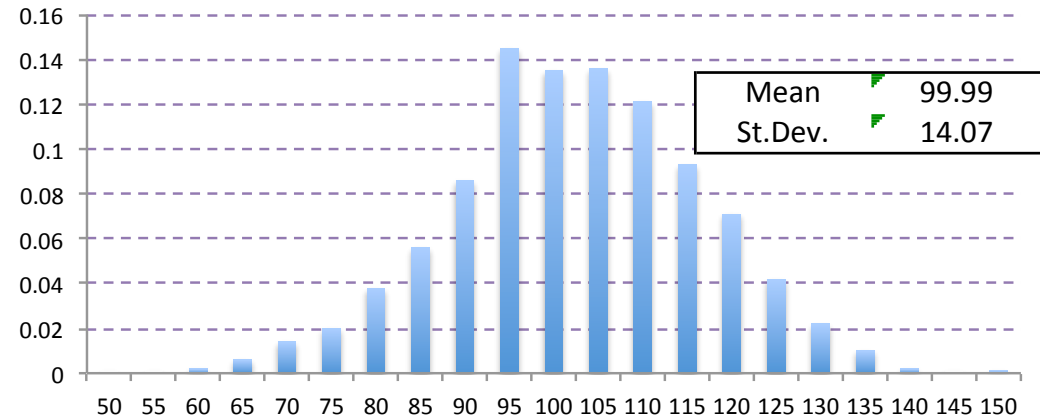


Acting without Overreacting

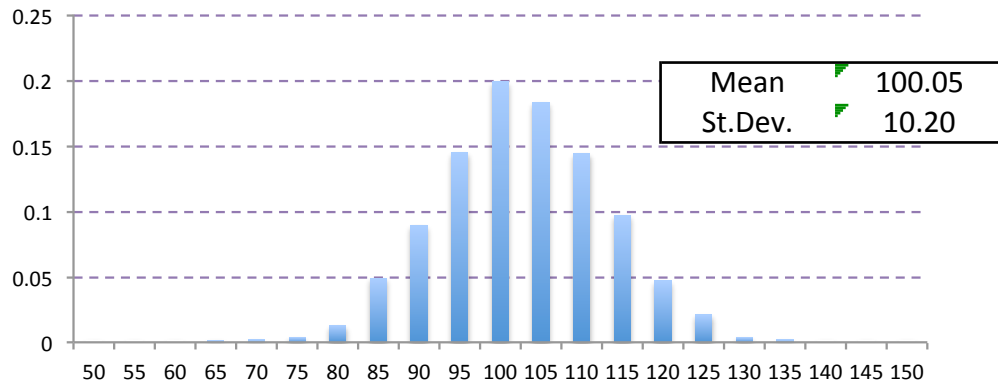
Target: $x^* = 100$

Variability: $\sigma = 10$

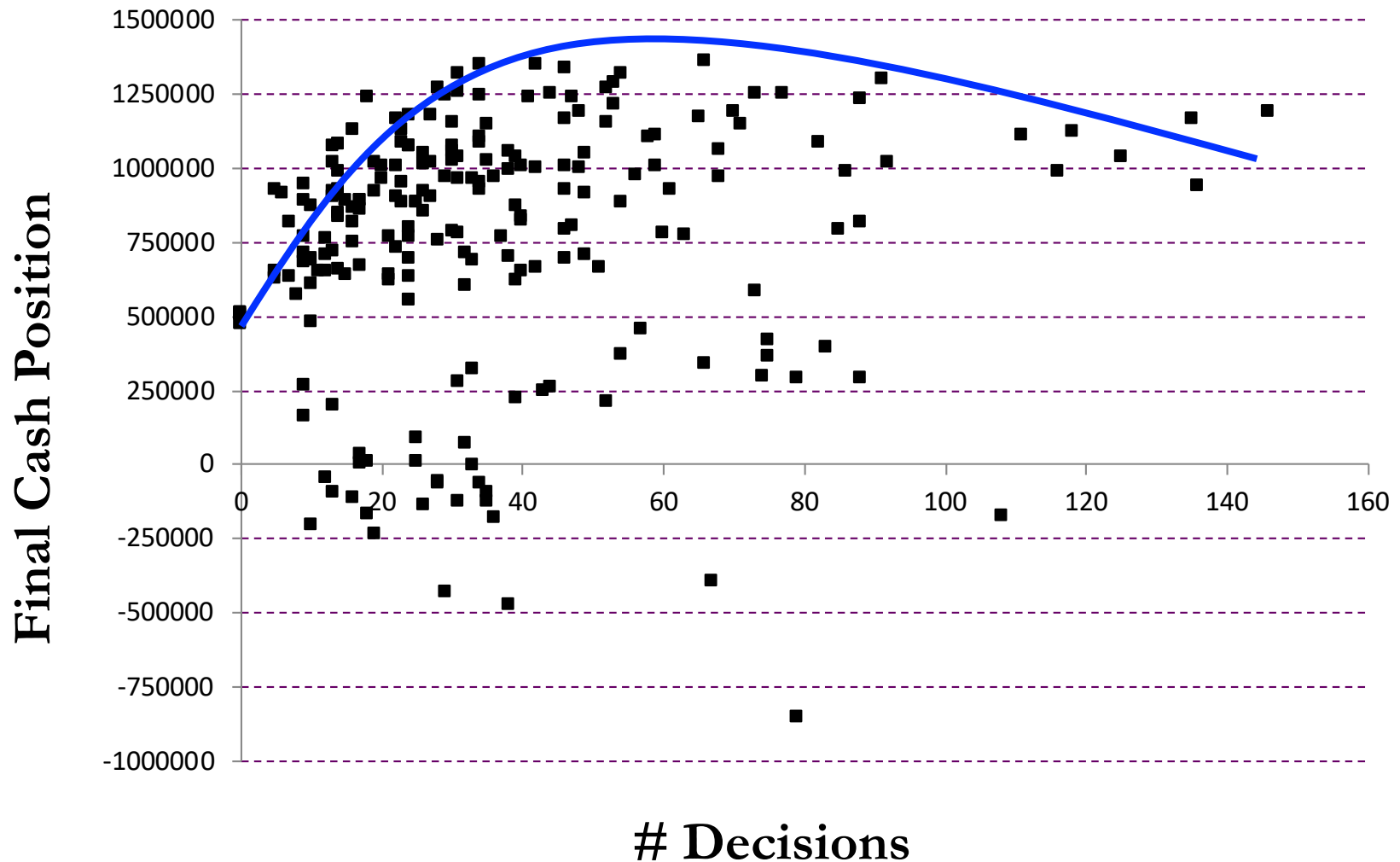
1st Idea: Adjust



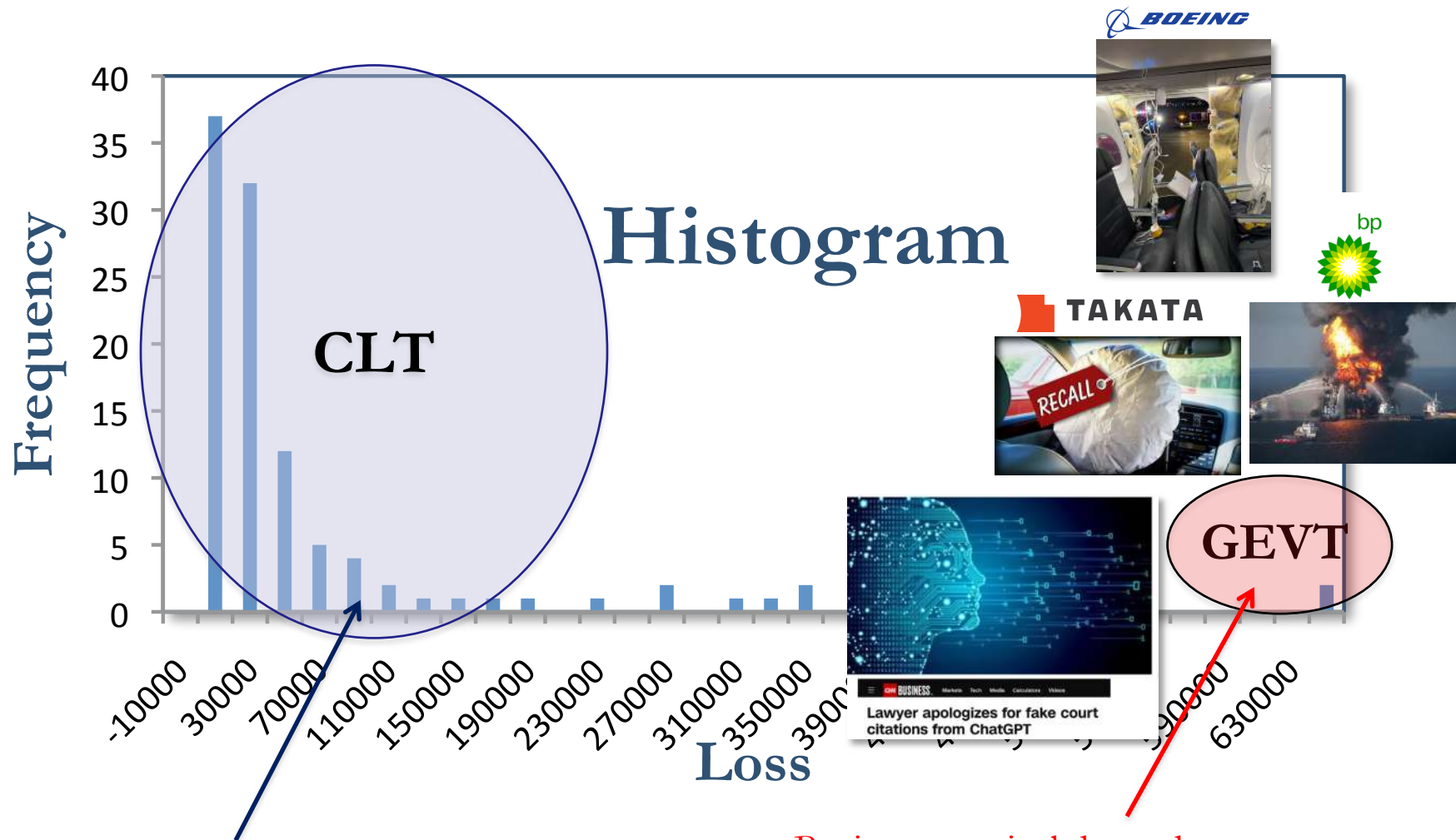
2nd Idea: Do not Adjust



Returns to Micromanagement



The Limits of SPC: Risk Lives in the Tail



Traditional SPC reduces chronic inefficiencies.

Business survival depends on managing low-frequency, high-severity tail risks.

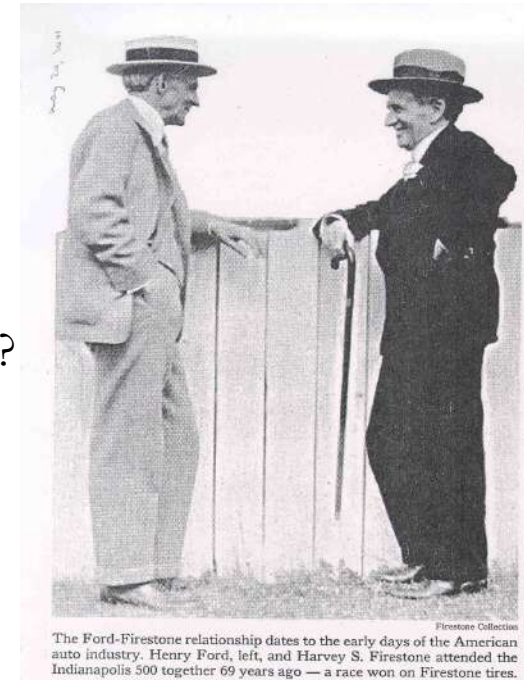
Ford-Firestone Case: 2000 Recall

➤ RECALL COST

- 6.5 million tires at \$50 (\$325 mil) + legal +liability +lost profits
- = \$800 million to \$1 billion
- Brand Image
- Most profitable segment for Ford
- 40% of sales for Bridgestone.
- Decatur plant was closed.
-

➤ Was this a real crisis or an overreaction? Was a recall necessary?

- Fatality rates
- Air (1970) .3 / 100 million miles
- Air (1997) .06 / 100 million miles
- Passenger cars 1.6 / 100 mil. mi.
- All compact SUV's 1.3 / 100 mil mi
- Ford Explorer 1.0 / 100 mil mi
- 100 deaths for 3.5 million Explorers on the road from 1990 to 2000



- This doesn't look so bad. Based on overall performance, the recall was unexpected. So what are the problems?

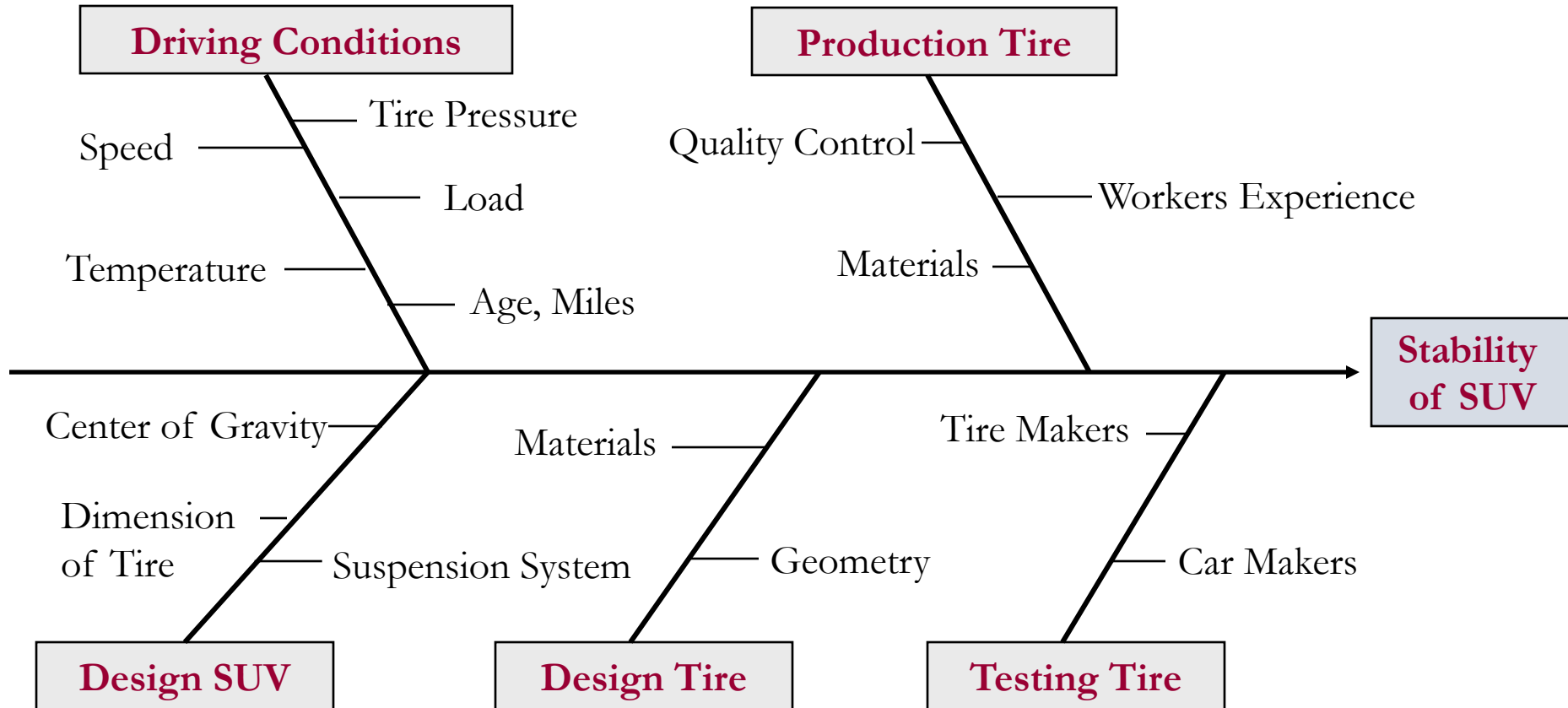
Timeline

- 1989-1990 (design stage), new tire launched (1996)
- 1997: Claims (?), Venezuela
- 1998: 21 cases, Saudi Arabia
- 1999: Ford memo to Firestone
- **2000 February: TV Show** 

**Tread separation
in Firestone tires
on Explorers**
- 2000 May: 90 complaints + 4 deaths
- 2000 August 9: issued recall notice for 6.5 million tires (went up to 13 million out of 47 million made in the 1990s)
- 2000 September: 2200 complaints + 103 deaths + 400 injuries

When should they have reacted? Ethics, Economics

Fishbone Diagram: Cause and Effect



Data: Ford and Firestone

Firestone argues that Ford Explorers are more accident-prone than Ford Ranger pickups equipped with the same tires...

...But Ford insists that Firestone tires are more likely to fail on Ford Explorers than are other tire brands

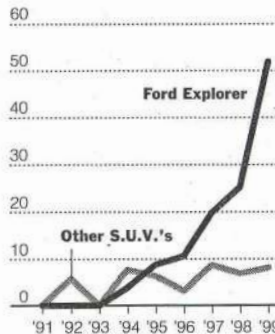


More Red Flags?

Federal data show that the rate of fatal accidents involving tires grew faster on Explorers than on other S.U.V.'s in the late 1990's and that the 1996 model year was particularly bad for these types of

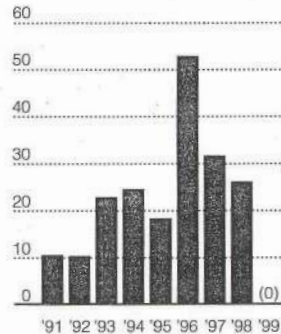
accidents. Other data from two Ford factories that build Explorers show that when either factory used Firestone tires instead of Goodyear tires, it had more tire-related fatalities.

Tire-related accidents per 1,000 fatal accidents; includes 1991 and later models

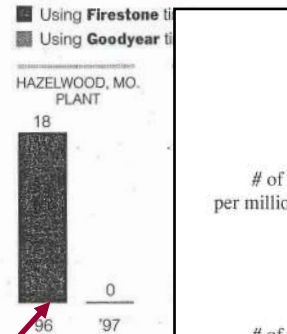


Sources: New York Times analysis of National Highway Traffic Safety Administration fatal accident data; Ford Motor Company

Tire-related accidents per 1,000 fatal accidents in Ford Explorers by model year



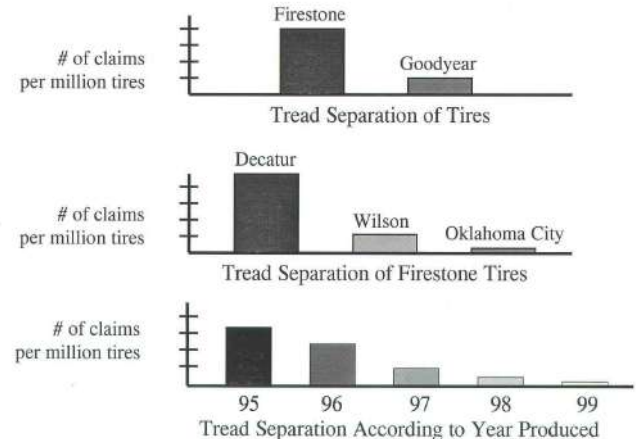
Tire-related fatal accidents involving Ford Explorers built at each factory by model year



Isolating causes by controlling for other variables.

Pareto Charts

Causes of Rollovers of Ford Explorers



The Risk Equation



$$\text{Risk} = \text{Detection} + \text{Frequency of Occurrence} + \text{Severity}$$

Acceptable risk levels:

Mother Nature

Random

Customers willing to take risk

Variation inherent in process/product

Unacceptable risk levels:

Assignable problem

Cause known but did not get caught

Now Mgrs. are responsible

They didn't do anything

when they found the problems!

What would customers say about the problem ?

Is the product too risky?

Are you doing enough to detect deviations?

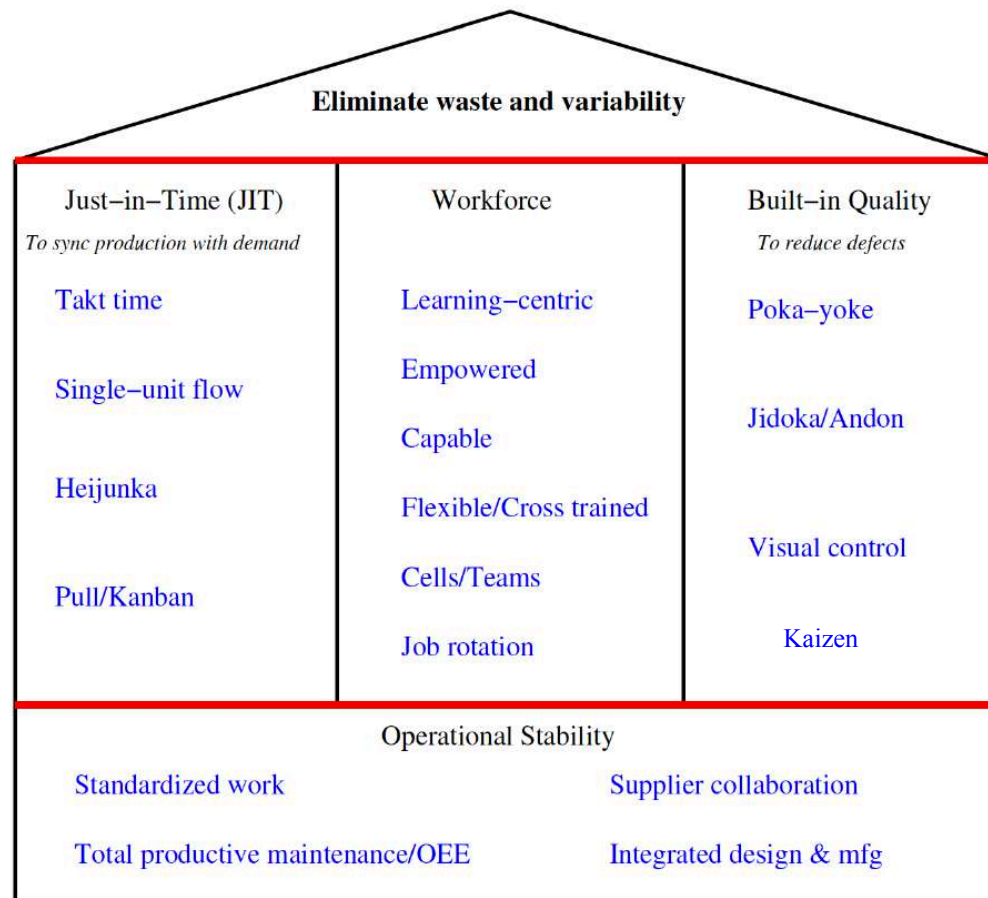
Total Quality Management (TQM)

Using the quality of products and services to increase a company's share of the market

- **Vision:** TQM stresses a commitment by management to have a continuing company-wide drive toward excellence in all aspects of products and services that are important to the customer. Encompasses entire organization, from supplier to customer.
- **Principle:** Quality is built in the processes.
- **Implementation:** Eliminate waste (lean). Everything that does not add value to the customer.

Toyota Motor Manufacturing

Lean Operations and the Toyota Production System





Available on YouTube [[Link](#)]

Toyota's Seat Fiasco



- What is this case about?
- What are the issues with the seat installation?
- What should Doug Friesen do?

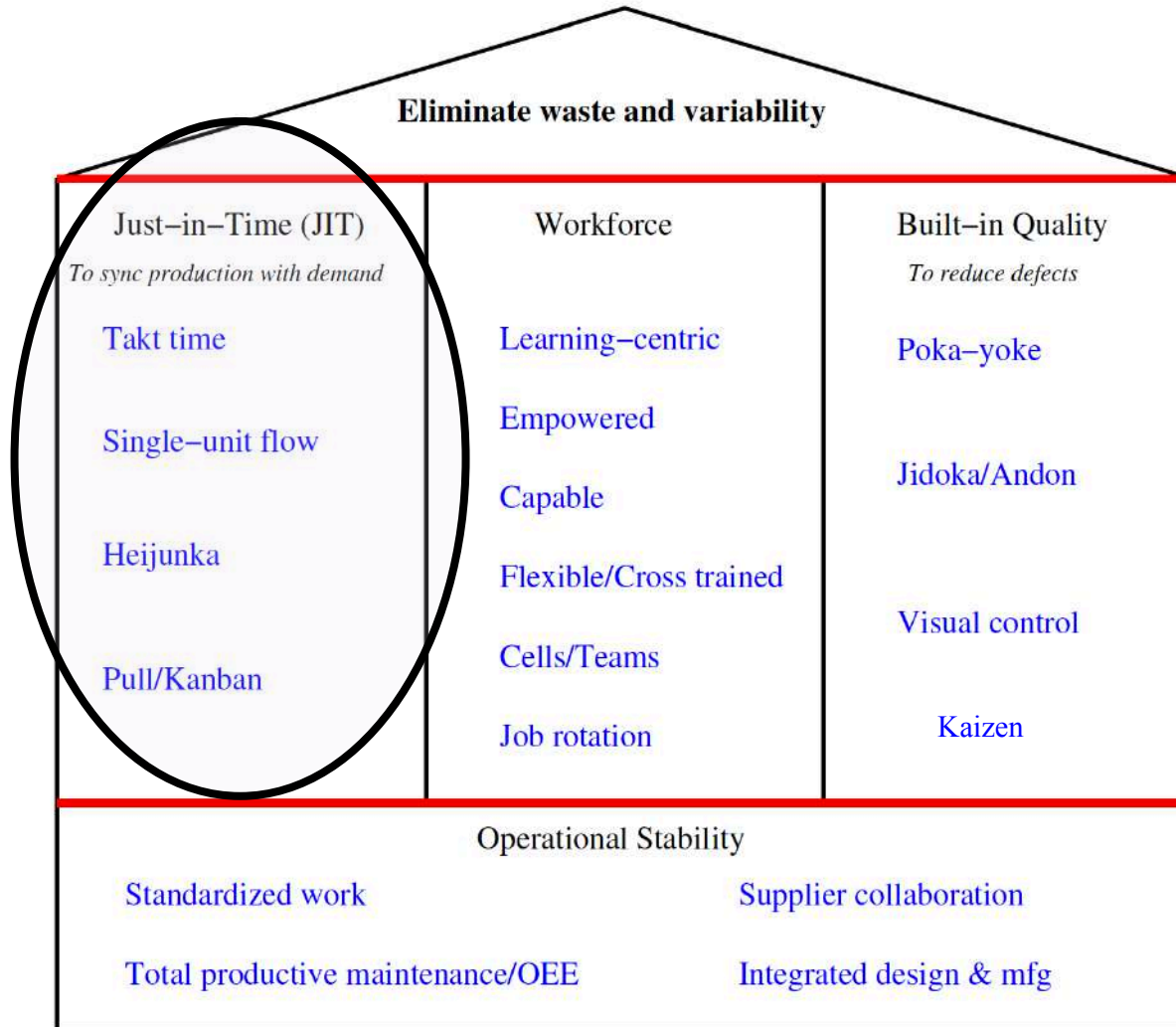
TPS Principles

➤ The Right Process Will Produce the Right Results

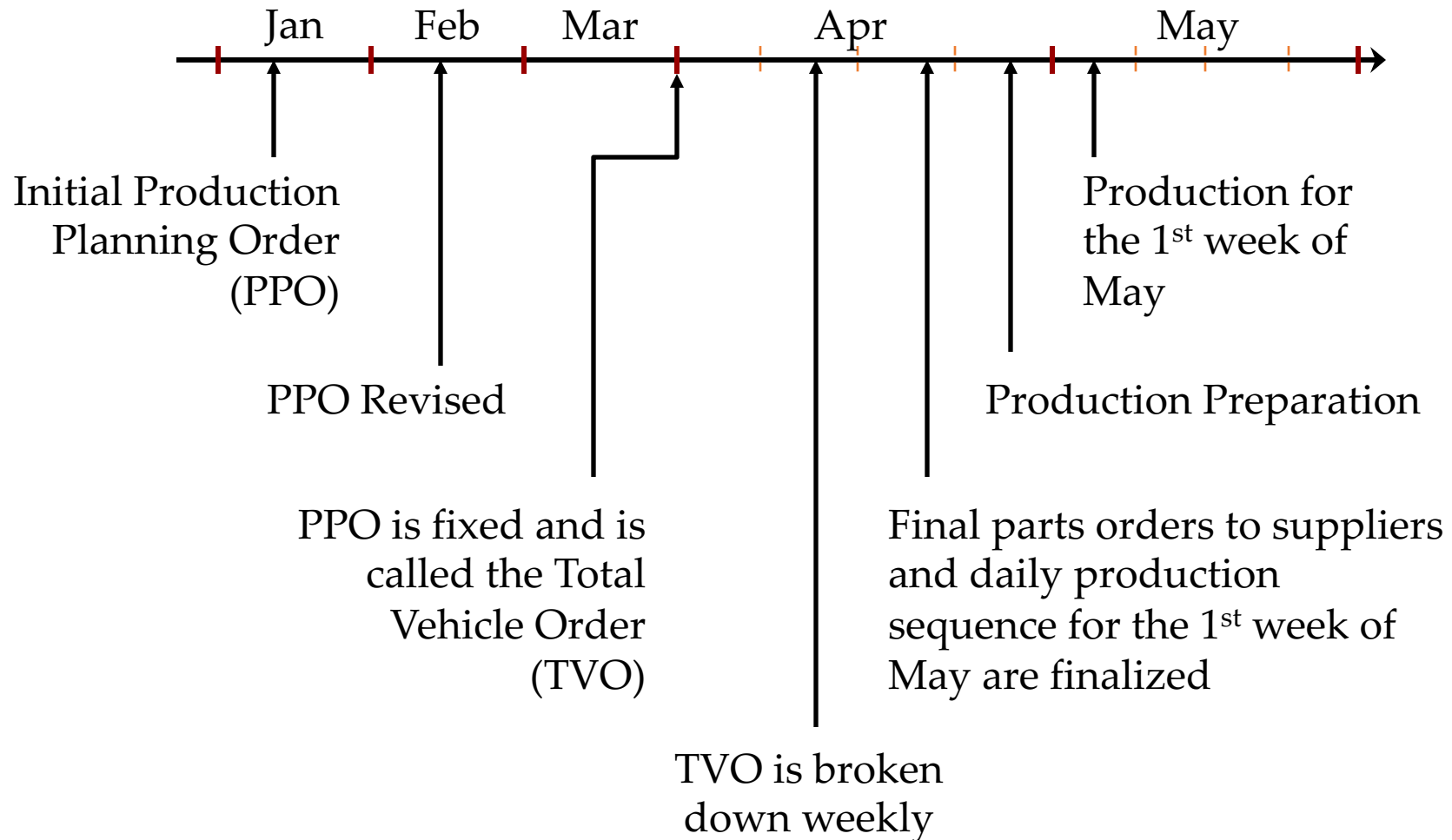
- ✓ **Principle 1:** Bring problems to the surface
- ✓ **Principle 2:** Use “Pull” system to avoid overproduction
- ✓ **Principle 3:** Level out workload
- ✓ **Principle 4:** Build a culture of stopping to fix problems
- ✓ **Principle 5:** Standardized for continuous improvement
- ✓ **Principle 6:** Use visual control so no problem is hidden
- ✓ **Principle 7:** Develop exceptional people and teams

TPS Framework: House of Lean

Lean Operations and the Toyota Production System



JIT Production: Production Control

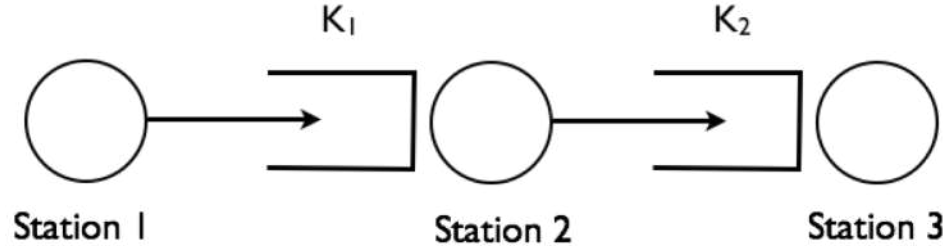


JIT production: Kanban Cards



- Every parts container carries a Kanban card including
 - part code #, description, batch size, delivery address, ...
- Card physically travels with the container and then back to the supplier.
- Pull production: only a Kanban triggers the production of parts.
- Information flow is embedded in the process.
- Inventory control: amount of WIP inventory is closely linked to # of Kanbans.
- Actual daily production sequence is given by Kanban cards, not heijunka.

Push vs. Pull Philosophy



K_1 and K_2 are bounds on the size of the buffers.

➤ Case 1: $K_1 = K_2 = 0$ (No buffers)

Starvation and blocking propagates immediately.

Minimal WIP.

PURE PULL

➤ Case 2: $K_1 = K_2 = \infty$ (Infinite buffers)

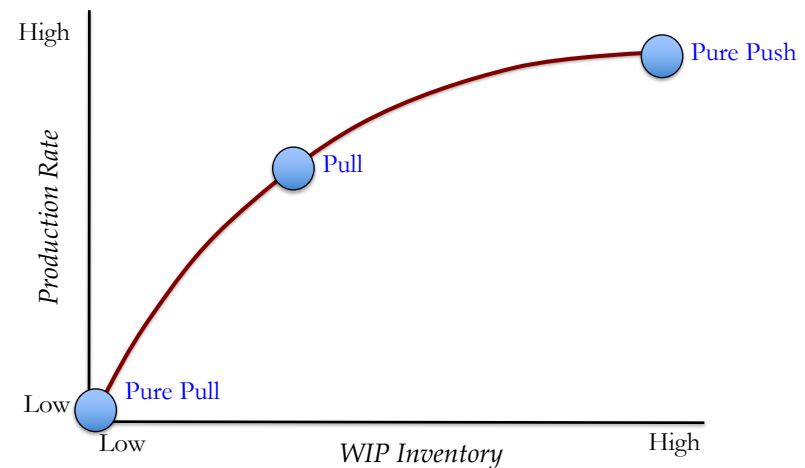
Delayed starvation propagation. No blocking.

Maximal WIP.

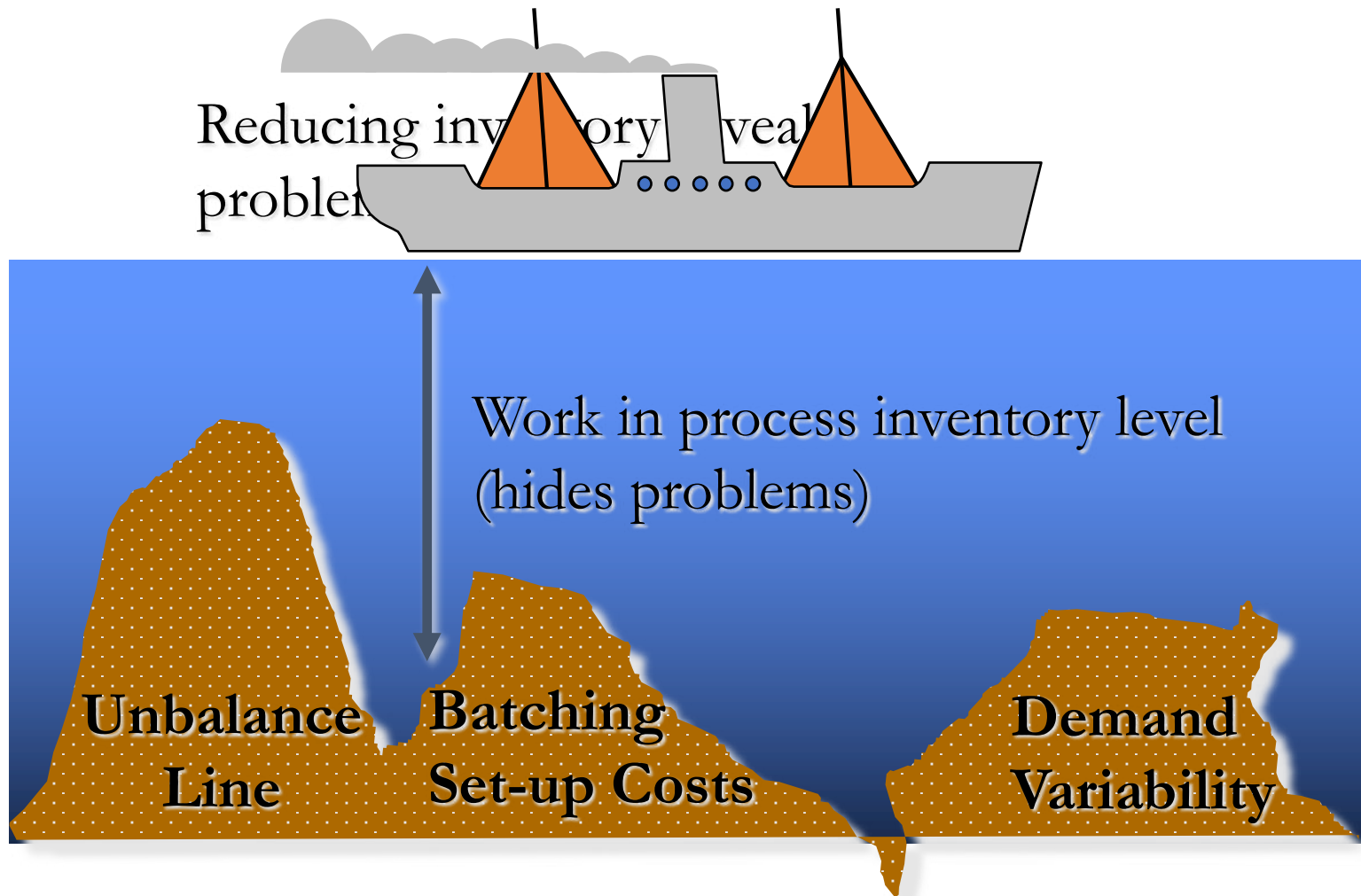
PURE PUSH

➤ Case 3: $K_1, K_2 > 0$ and not too large

- ✓ K_1 and K_2 equal the number of kanbans between the segments.
- ✓ Available kanban = an empty buffer slot = “authorization to work.”
- ✓ Blocking and starvation are mitigated with small amounts of WIP.
- ✓ Forced idleness cascades over time.

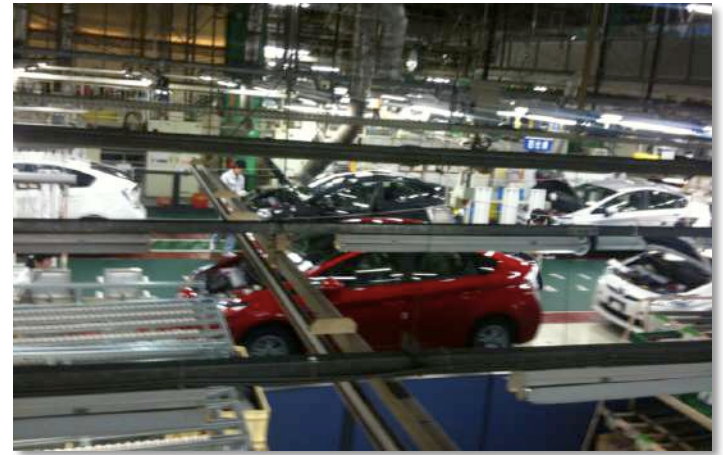


Implementing JIT: Inventory Analogy



Implementing JIT: Heijunka

- Evening out (balancing) the total order in the daily production sequence.
- Variety in May 1992: 23 sedan and wagon models, 11 exterior colors, 29 interior variations & 30 other options.
- Benefits
 - Level production
 - Less risk of inventory obsolescence
 - Synchronization with sales
- Challenges and complexities
 - Reduce setups to allow frequent changeovers
 - Incorporate production schedule in material and information flows



Ford Model T

“Any customer can have a car painted any color that he wants so long as it is black.”

Henry Ford



Honda has some of the most flexible plants in the U.S. To illustrate that flexibility, Honda is able to switch from producing Civics to CR-Vs with only 5 minutes of downtime!

“Honda’s Flexible Plants Provide Edge” WSJ, 09/23/2008

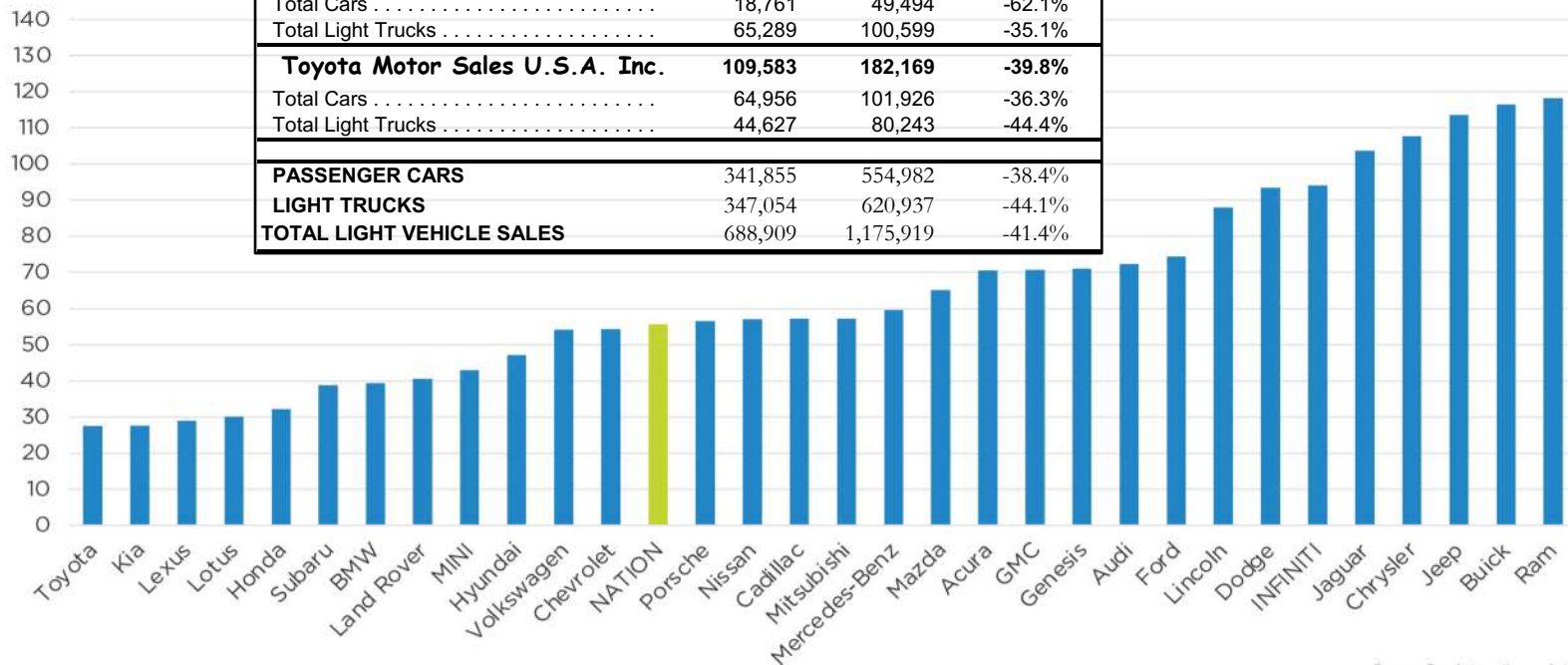
Benefits of Heijunka



U.S. Light Vehicle Retail Sales - February 2009

	Feb 2009	Feb 2008	%Chng.
General Motors Corp.	126,170	268,737	-53.1%
Total Cars	53,813	107,592	-50.0%
Total Light Trucks	72,357	161,145	-55.1%
memo: Saab.....	712	1,734	-58.9%
Ford Motor Company	99,050	192,178	-48.5%
Total Cars	36,765	63,270	-41.9%
Total Light Trucks	62,285	128,908	-51.7%
memo: Volvo.....	3,356	7,505	-55.3%
Chrysler LLC	84,050	150,093	-44.0%
Total Cars	18,761	49,494	-62.1%
Total Light Trucks	65,289	100,599	-35.1%
Toyota Motor Sales U.S.A. Inc.	109,583	182,169	-39.8%
Total Cars	64,956	101,926	-36.3%
Total Light Trucks	44,627	80,243	-44.4%
PASSENGER CARS	341,855	554,982	-38.4%
LIGHT TRUCKS	347,054	620,937	-44.1%
TOTAL LIGHT VEHICLE SALES	688,909	1,175,919	-41.4%

Days of Supply



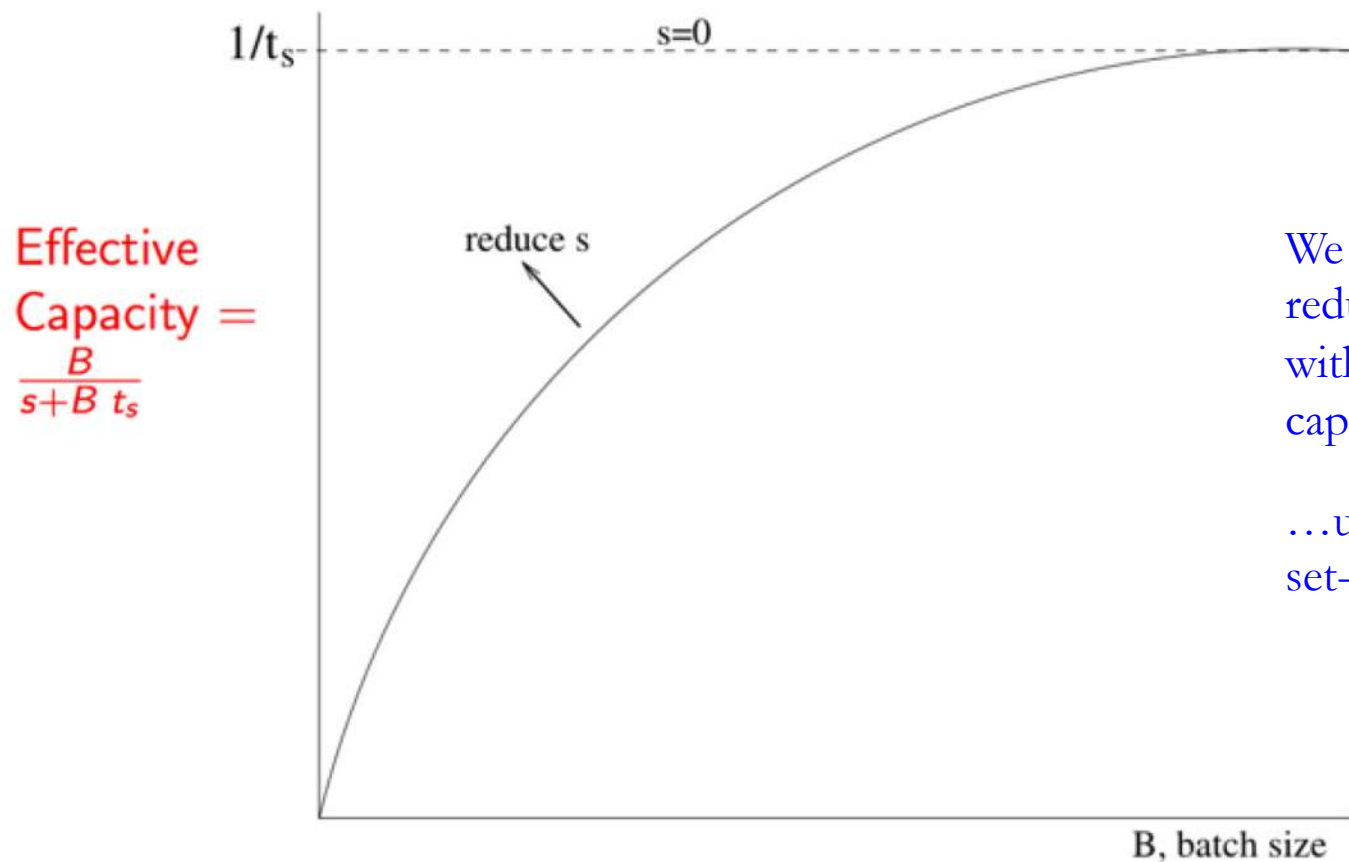
Source: Cox Automotive - vAuto

Note: Days' supply calculation includes vehicles in dealer inventory and in-transit/pipeline

Challenges to Heijunka

Let B denote the batch size, s the setup time, and t_s the unit processing time.

The time to produce a batch of size B is $s + B \times t_s$

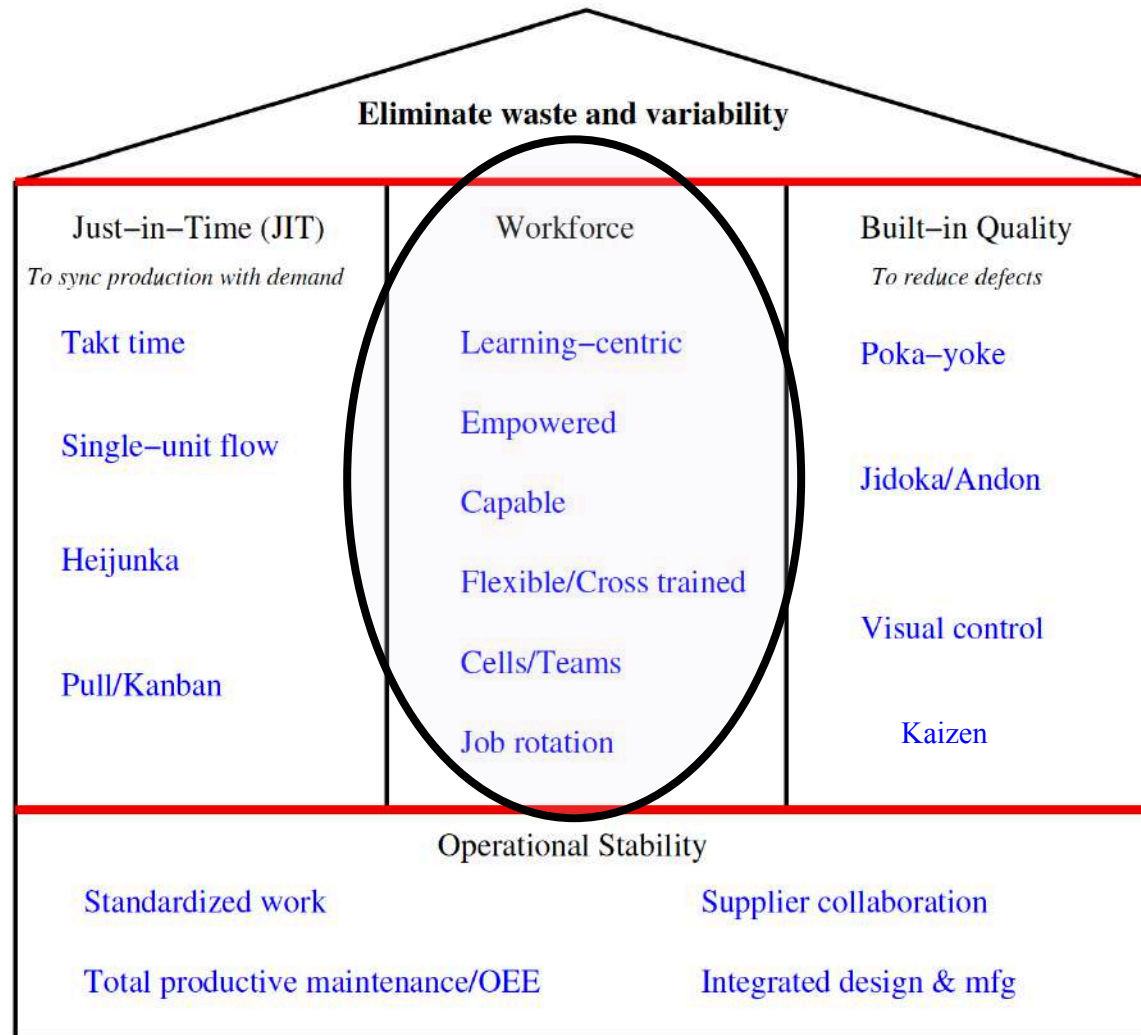


We cannot simply reduce batch sizes without sacrificing capacity

...unless we eliminate set-up times.

TPS Framework: House of Lean

Lean Operations and the Toyota Production System



People



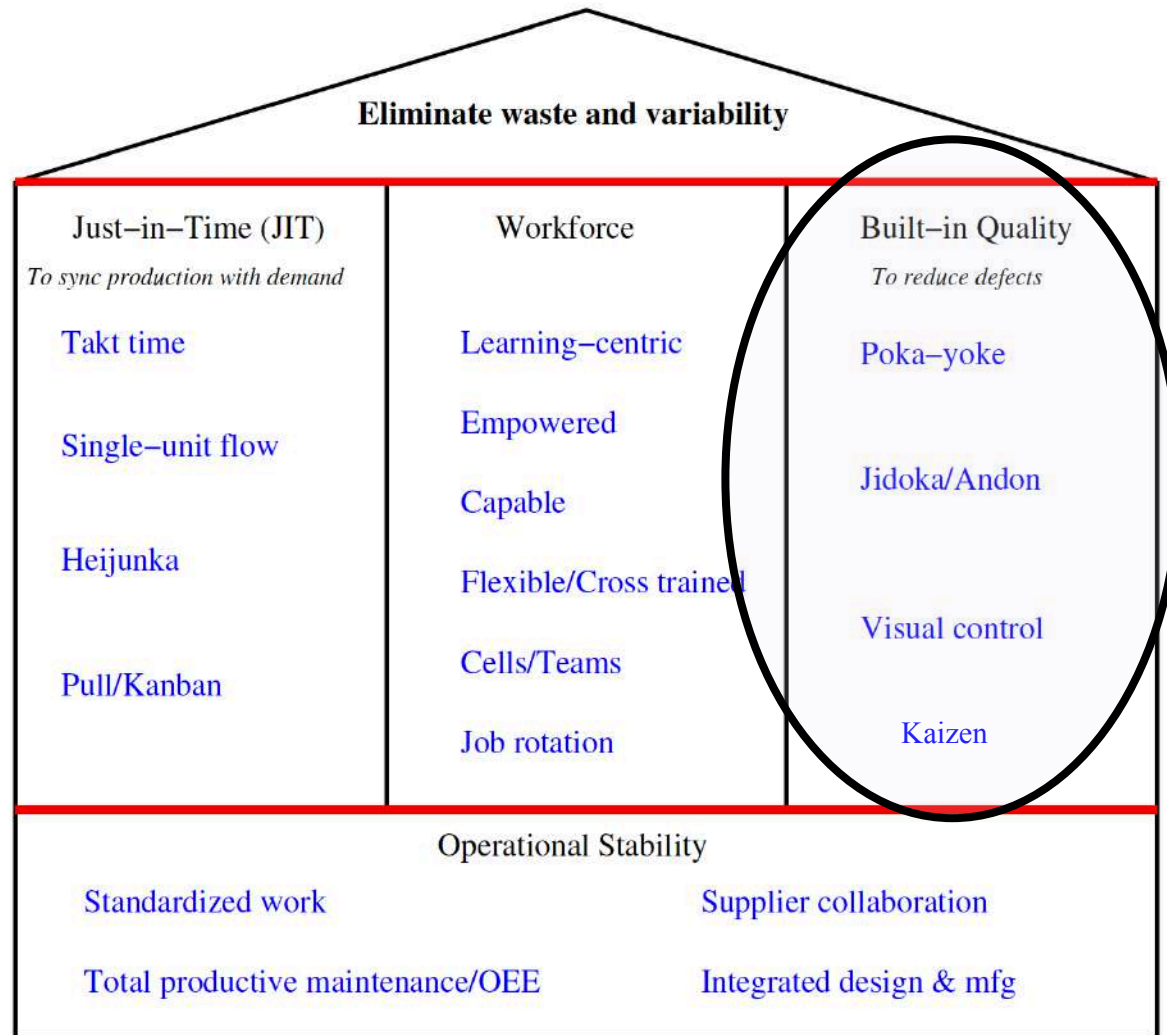
- Patiently select and invest in people's knowledge and skills
 - One-on-one coaching
 - Learn by doing: *continuous improvement*
- Worker: training and discipline to add value and improve processes; solve problems by root cause analysis and countermeasures (*Employee Empowerment*)
- Team Leader: most knowledgeable/skilled in work zone; best leader-teacher.
- Group Leader: Experience as a team leader plus broader system knowledge
- Assistant Manager: Broader skills and knowledge
- Role of managers: strategic thinking, not fire-fighting.

“To teach [TPS] well, you have to get to know people very deeply and over time. In the process we all become students here.”

- Mike DaPrile, Assembly Plant Manager,
 - Toyota Motor Manufacturing,
 - Georgetown, Kentucky, Plant.

TPS Framework: House of Lean

Lean Operations and the Toyota Production System



TPS Principle: Jidoka

- **Make any production problems instantly self-evident and stop producing whenever problems are detected**
 - Build in quality in the production process
 - Any deviation from value addition is a waste
 - Value is defined from the viewpoint of the next station down the line
- **Standardize the process and document standards**
 - Work chart: cycle time, sequence of tasks, breakup of cycle time into the tasks
 - Colored tape to mark where everything belonged
- **Andon cord**
 - Why detect problems when and where they happen?
 - What are the costs and benefits of pulling the andon cord?
- **Andon board**
- **Highlight problems to successively higher levels, e.g., Code 1 status**

Andon System

(The cost of stopping the line)

Question: If the line stops for one cycle (57sec) then the cost would be

1. The value of a car \$18,500.
2. The profit margin per car 17% of \$18,500 $\approx$ \$3,145
3. The variable (labor) cost of a cycle $\approx$ \$0.269 per worker

- Facts:
 - 12 andon pulls per team member with 1 actual stoppage
 - Standard time per shift = 450 minutes
 - Run ratio = 85-95%
- Lost production per shift = 5 to 15% of 450 minutes = 22.5 to 67.5 minutes.
- Consider Final 1 with 30 people
 - # of cord pulls = $30 \times 12 = 360$
 - # of stops per shift (1 in 12) = $360 / 12 = 30$ stops
 - Avg. downtime per stop = $\text{Downtime} / \# \text{ stops} = [22.5 / 30, 67.5 / 30] = [0.75, 2.25]$ mins/stop
- Cost:
 - OT / person-min = $(\$17 / 60) \times 1.5 = \0.425 per minute.
 - Final 1 - OT/min = $30 \times 0.425 = \$12.75$ per minute.
 - Total Cost = $\$12.75 \times [22.5, 67.5] = [\$287, \$861]$ per shift

Cost of stopping and not stopping the line

➤ **Cost of stopping the line:**

- The variable (labor) cost of a cycle: \$0.269 per worker

➤ **Cost of not stopping the line:**

- Need for rework
- Propagation of defective products through the system
- Excessive waste
- ...
- Difficult to attach a \$ number to this phenomenon

➤ **Toyota Philosophy:**

- the risk of snowballing waste is such that stopping the line is preferred to not stopping the line even if the cost of not stopping is not quantifiable.

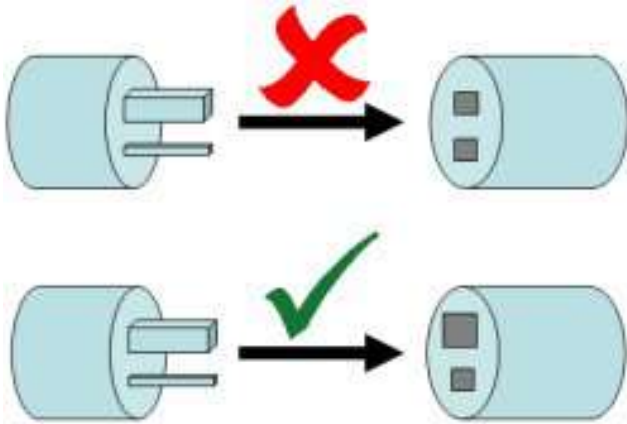
TPS Principle : Kaizen

- “changing something for the better”
- Eliminate **waste** in seven categories



- How does Quality Control work at TMM
 - **Eliminate fear**: Separate the people from the problem.
 - **Root cause analysis**: 5 Why's

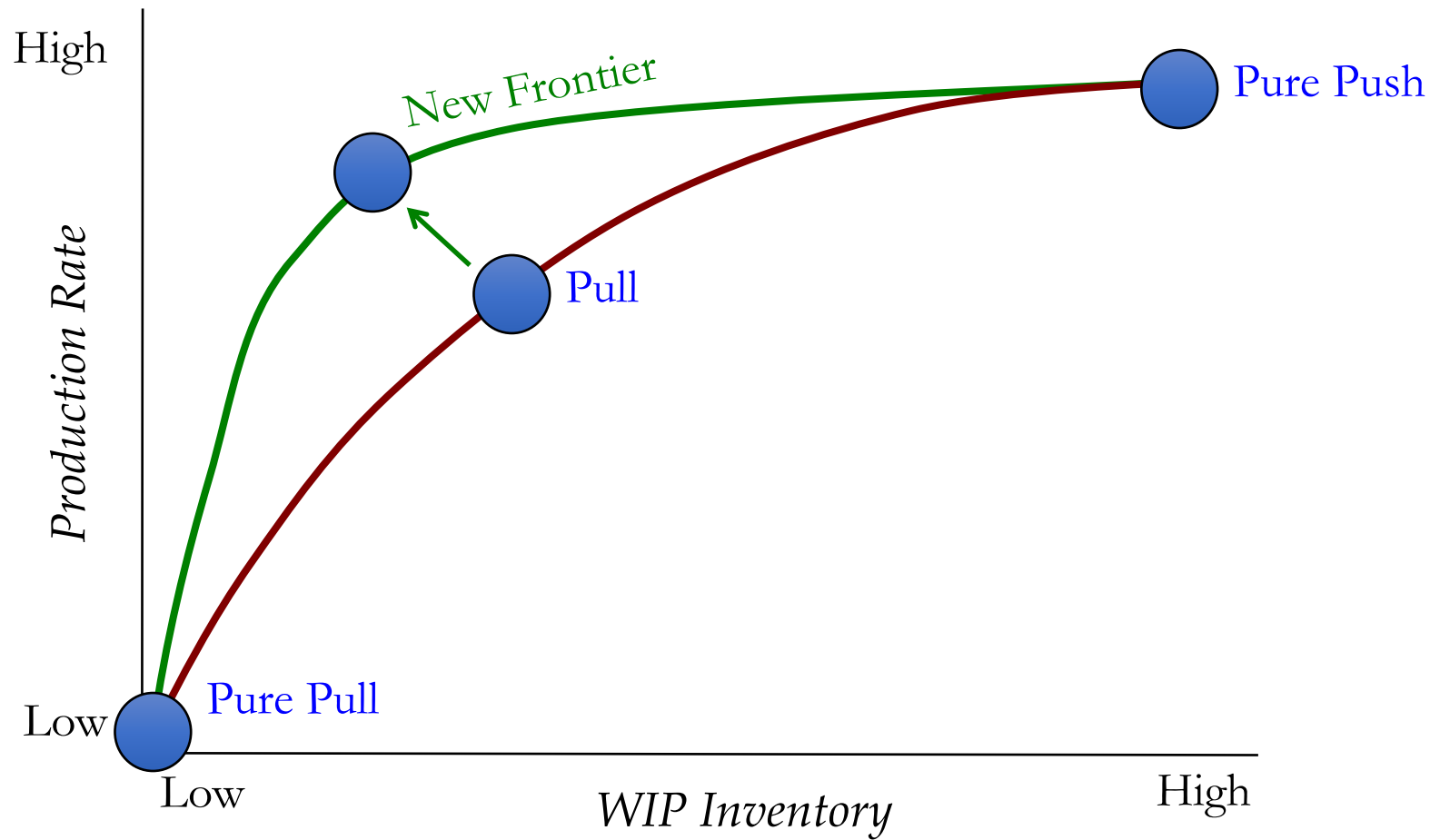
Mistake Proofing (Poka-yoke)



Hospital Door Handle That Sanitizes Hands With a Touch



TPS Principle: Continuous Improvement



Six Sigma

(Quality as “Conformance to Specifications”)

Any variation from our target is considered a defect

- “The Application of rigorous quantitative tools in the planning, control and improvement of processes in a effort to meet customer requirement 100% of the time”
- Involves every part of the company
 - People, equipment, suppliers, materials, procedures...
- Other names: “zero-defect”, Kaizen
- Pioneers: Motorola, GE

Objectives: Eliminate Defects and Process Variability

- Tool for Continuous Improvement: Prevention is More Effective than Inspection
- Tool for Risk Mitigation: Prevention is More Effective than Correction

Six Sigma Metric

- ✓ Count the defects
- ✓ Express them as defects per unit (DPU)
 - Unit: An item being processed, or the final product or service being delivered to the customer
- ✓ 52 out of 250 loan applications contain defects
 - $52/250 = 20.8\%$ DPU
 - This measure sometimes too crude for two reasons

Reason 1: Quest for Zero-defects

- Federal Express ships 99.999% of all orders correctly
- Motorola has achieved 3.4 defects per million; a 100 fold improvement since 1981. It is moving towards 60 defects per BILLION.
- Kodak had 2 defects in 600 million plastic tips used in its blood analyzer machines (the packages were not closed correctly! They did not reach the customer)
- There is no end to the quest for quality
- Express in DP*million*U

Reason 2: Apples & Oranges

- **Complex products have greater “opportunities” for making a mistake**
 - ✓ Make metrics comparable across processes
 - ✓ List defect opportunities
 - ✓ Determine which are actual, customer critical, specific defects
 - ✓ Check the proposed number of defect opportunities against other standards (industry, govt., firm, competition, other similar products/services)

- **Example: 52 defects, 250 loan applications, 4 defect opportunities per application**
 - ✓ **Defects per Opportunity** = $52 / (250 \times 4) = 0.052 \text{ DPO}$

Views of Quality

Classical View of Quality *"99% Good" (3.8σ)*

- ➡ 20,000 lost articles of mail per hour
- ➡ Unsafe drinking water almost 15 minutes each day
- ➡ 5,000 incorrect surgical operations per week
- ➡ 2 short or long landings at most major airports daily
- ➡ 200,000 wrong drug prescriptions each year
- ➡ No electricity for almost 7 hours each month

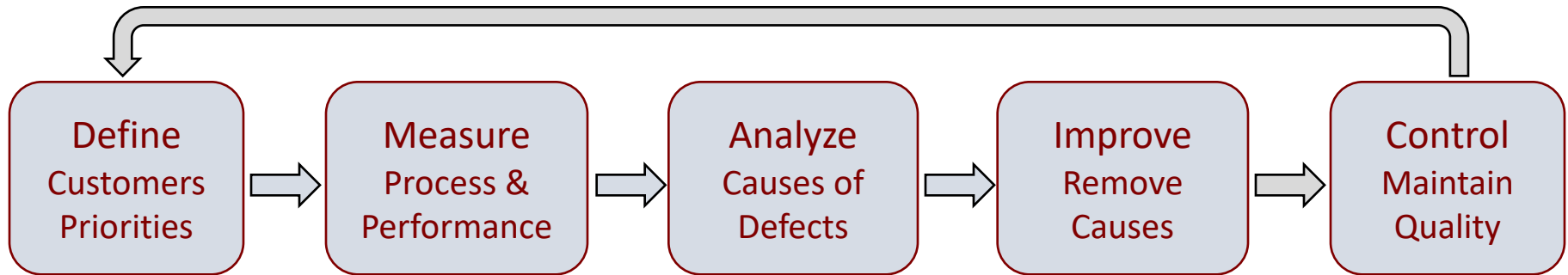
Six Sigma View of Quality *"99.99966% Good" (6σ)*

- ➡ Seven lost articles of mail per hour
- ➡ One minute of unsafe drinking water every seven months
- ➡ 1.7 incorrect surgical operations per week
- ➡ One short or long landing at most major airports every five years
- ➡ 68 wrong drug prescriptions each year
- ➡ One hour without electricity every 34 years

Continuous Improvement: DMAIC Cycle and Six Sigma



Continuous Improvement



- Process Flow Map
- Benchmark
- Voice of the Customer
- Quality Function Deployment
- Gap Model

- Data Collection
- Sampling Techniques
- Defect Metrics

- Fishbone Diagram
- Sampling Techniques
- Influence Diagram
- Root Cause Analysis (5 Whys)
- Pareto Charts
- Control Charts
- Regressions
- Capabilities

- Design of Experiments
- Kaizen
- Standardization
- Lean Ops
- Robust Design
- Poka-Yoke
- Training

- Andon System
- Preventive Activities
- SPC
- X-charts
- P-charts

....Back to the Seat Problem

Actions Taken by TMM and KFS after May 1 (not in the case)

➤ At KFS:

- Seat reorder form was revised.
- KFS: altered lighting in their final inspection area
 - Insufficient lighting could have been a problem at KFS inspection area

➤ At Toyota:

- KFS stationed QC engineers at TMM seat receiving area
 - To discover more about the problem
- Clearly marked overflow area for seat-defective cars
 - When this space was filled, a mini-May 1 type meeting would be held to give immediate feedback to KFS.
- Bulletin board in overflow area with 8 slots for different kinds of defects and a copy of the reorder form was placed in these slots:
What defects are common? How fast are replacements arriving?

....Back to the Seat Problem

- More kaizen training to group and team leaders.
- Training for KFS.
- By the end of May 1992: heightened awareness of seat defects, faster problem-solving cycle
- Results: weekly defect counts fell by 25 %
- Continued improvement even after the introduction of several new models in the summer.
- **Summary of the seat problem:**
 - Managerial Problem stemming from the lack of a system that helped people autonomously improve the status quo.
 - Such problems will always occur in a dynamic environment and a relentless pursuit of the cause of the defects and opportunities for improvement is necessary.

“the good things I saw [at Tsutsumi] were just common sense and no big deal at all. My eyes weren’t open back then.”

- Doug Friesen, Manager of Assembly,
 - Toyota Motor Manufacturing,
 - Georgetown, Kentucky, Plant.

More on the Toyota Production System

- Toyota: has been intensively researched and documented
- Practices (kanbans, for example) have been copied in many other companies.
- Many companies have tried to develop their own versions of the TPS and have failed.
 - Reason: Not cultural.
 - Even Nissan and Honda could not
 - TPS has been introduced successfully in Toyota, North America.
- Why is it so difficult?
 - TPS creates a community of scientists.

“Decoding the DNA of the Toyota Production System” by Steven Spear and Kent Bowen
Harvard Business Review (available on course website).

Other Applications of TPS

- **ZARA** uses TPS production and logistics to get new designs into store in two weeks while keeping costs low.
- **New Balance** TPS program was able to cut the flow time of production of a pair of shoes from 8 days to 2 hours. (The actual work content is only 23 minutes).
- **Marriott** has applied lean to its operations (The Novi, MI, Marriott Towne Place Suites) raising its services 10 points to 87 and went from the bottom half to the top 10% in its peer group.
- **Virginia Mason Medical Center** has pioneered the use of TPS in Health Care. For example, VMMC reduced cumulative nurse walking distance in the hospital by 750 miles per day, which freed up more than 250 hours of time for direct patient care.

Key Ideas

➤ Statistical Process Control

- SPC is an early-warning system
- Separate common-cause vs special-cause variation
- Correcting without overcorrecting
- The limits of SPC: variations versus risks

➤ Risk = Detection + Frequency of Occurrence + Severity

➤ The Toyota Way: The House of Lean

- TQM/TPS/Lean Ops is a strategic choice of process mgmt.
- Just-in-time (JIT) – Pull Production – Kanban System - Heijunka
- Continuous improvement: Prevention is More Effective than Correction
- Employee empowerment: Separate the people from the problem
- Andon Systems and Jidoka - Make problems visible

After Class Activities

➤ INDIVIDUAL:

PREPARE FOR MIDTERM

Sample test available on Canvas

➤ GROUP:

PREPARE GROUP PROJECT REPORTS

Instructions on Syllabus and Canvas